



EE345 – 第三代半导体基础导论

Introduction to Wide-bandgap Semiconductors

Chapter 04



Review of last lecture

1. Planar structure MOS and FinFETs
2. Crystal growth of SiC
 - Acheson process
 - Lely process
 - Modified Lely process (Seeded sublimation method)
 - Solution-based epitaxy
 - High temperature CVD
3. Defects:
 - Micro-pipes
 - Dislocations
4. Crystal growth of GaN





Contents in this Chapter

1. Characterization methods





Characterization techniques

If I give you an electronic device, what kinds of measurement methods you will do?





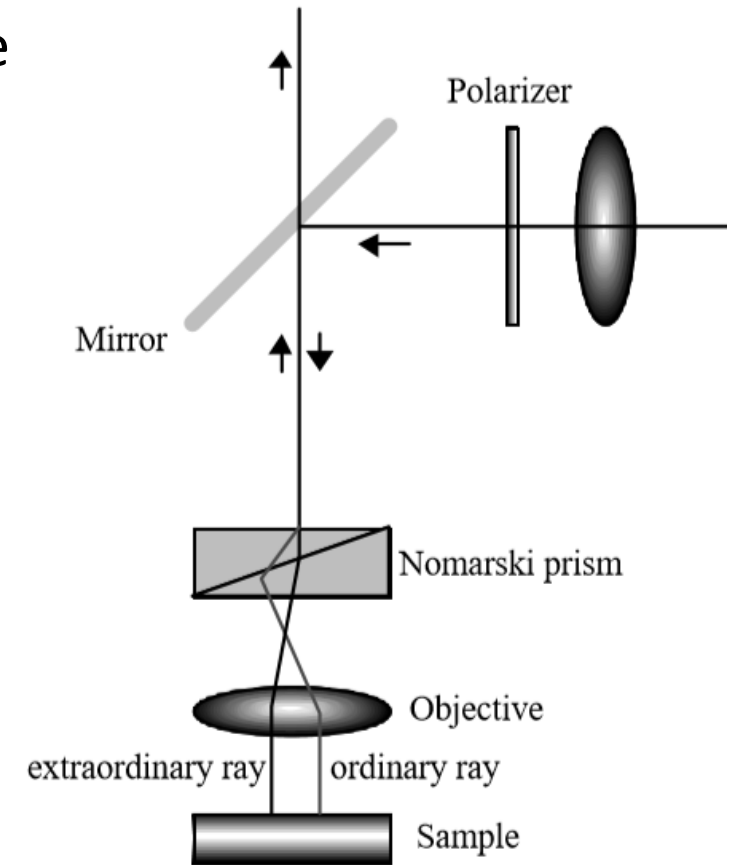
Characterization techniques of epitaxial SiC and GaN

- Optical microscopy with Nomarski interference contrast
- KOH etching
- Scanning Electron Microscopy (SEM)
- Cathodoluminescence (CL)
- Hall Effect measurements
- Atomic Force Microscopy (AFM)
- Transmission Electron Microscope (TEM)
- X-ray diffraction (XRD)
- Secondary ion mass spectrometry (SIMS)
- X-ray photoelectron spectroscopy (XPS)
- Mercury Probe C–V
- Photoluminescence (PL)
- Raman spectroscopy
- Ellipsometry



Optical microscopy with Nomarski differential interference contrast (DIC)

- The main principle of the Nomarski differential interference contrast (DIC) consists in the difference of the optical path.
- The Nomarski prism splits an incident plane light beam into two polarized light: an ordinary and an extraordinary ray
- After reflection on the sample, the light beam passes through the prism to an analyzer
- Height difference between objects (particles, defects ...) on the surface could be observe



Schematic principle of the Nomarski interference contrast



微分干涉相差显微技术

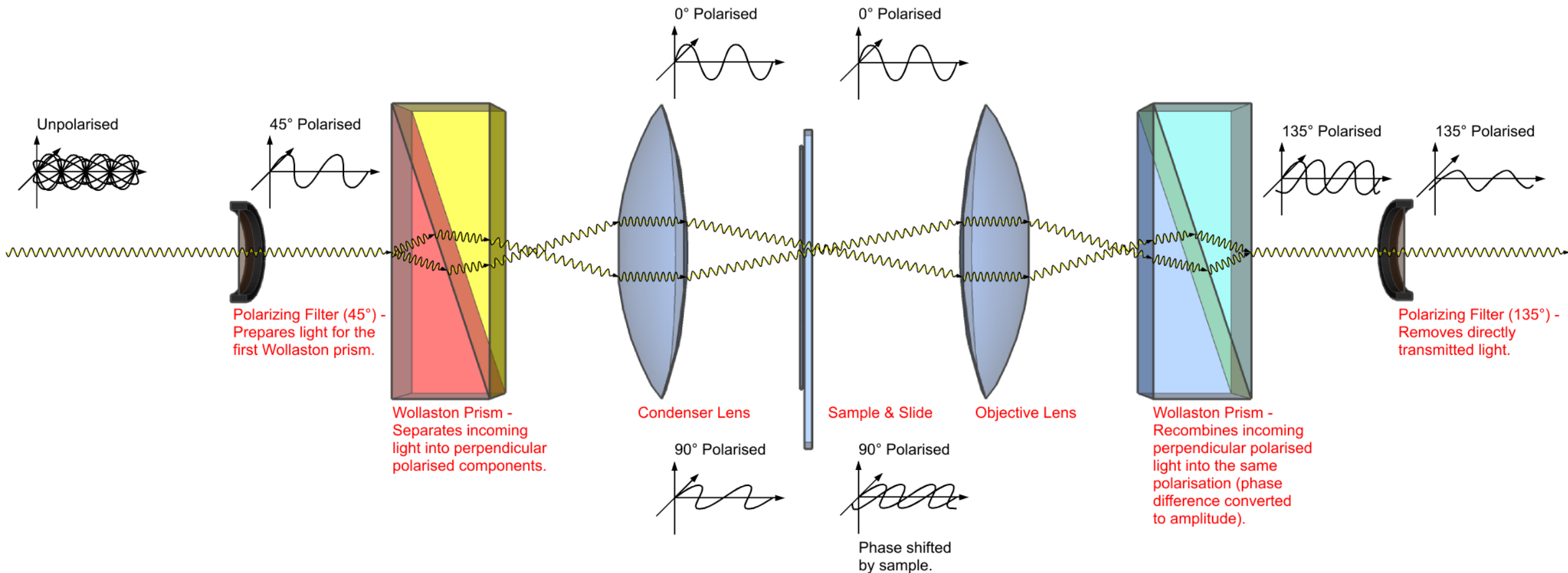
- 微分干涉相差显微技术(DIC), 又称**Normarski**干涉相差显微技术或**Normarski**显微镜, 是一种增强对比度来观察未染色的透明的样品的光学显微镜。DIC根据干涉测量获取有关样品光路长度（英语：Optical path length）信息，以查看其他不可见的特征。相对复杂的光学系统产生具有灰色背景的黑色或白色的图像。该图像类似于通过相差显微镜获得的但没有明亮衍射光晕的图像。该技术由波兰物理学家**Georges Nomarski**（英语：Georges Nomarski）在1952年研发。^[1]
- DIC通过将偏振光源分离成在样品平面上空间位移（剪切）的两个正交偏振相干部分，并在观察之前重组。复合时两部分的干涉对其光程差（即折射率乘积和几何路径长度）敏感。添加可调节的偏移相位确定在所述样品中的零光程差的干涉，对比度是正比于沿剪切方向的路径长度梯度，得到三维的光密度变化的样本图像，图像强调线条和边缘，但不提供表面上准确的图像。

From Baidu

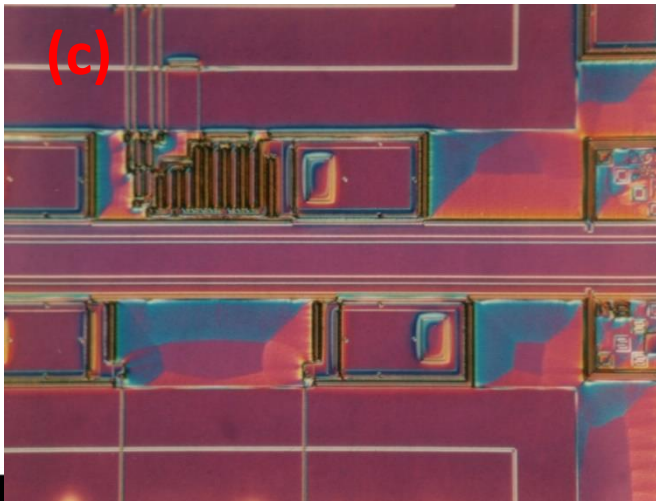
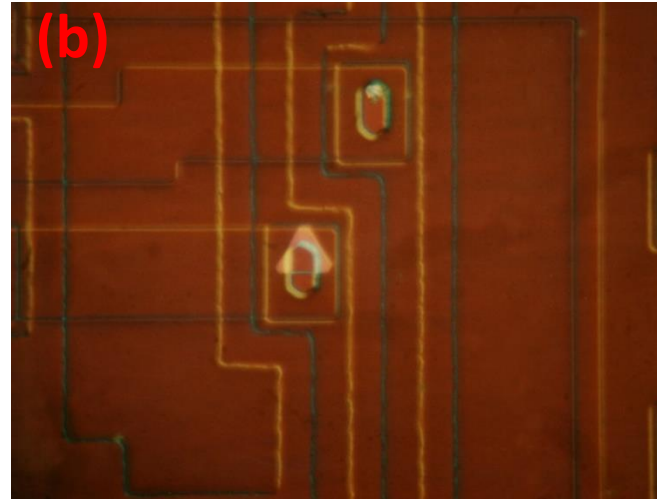


Nomarski DIC Microscopy

Nomarski Differential interference contrast (DIC) microscopy



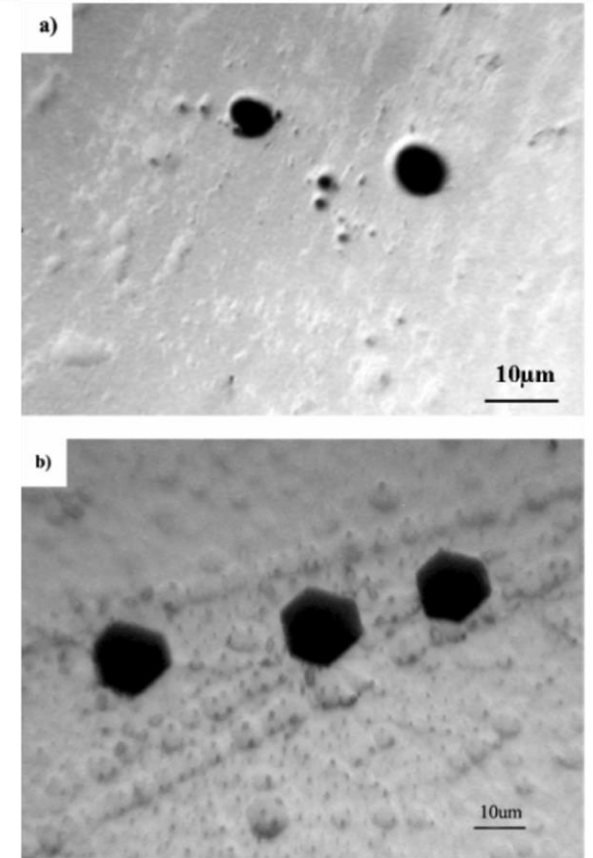
Typical results



- a) *Micrasterias furcata* imaged in transmitted DIC microscopy
- b) Aluminum–silicon alloying pit made visible by Nomarski DIC
- c) Partially developed photoresist in Nomarski DIC
- d) Partially etched silicon dioxide in Nomarski DIC

KOH Etching

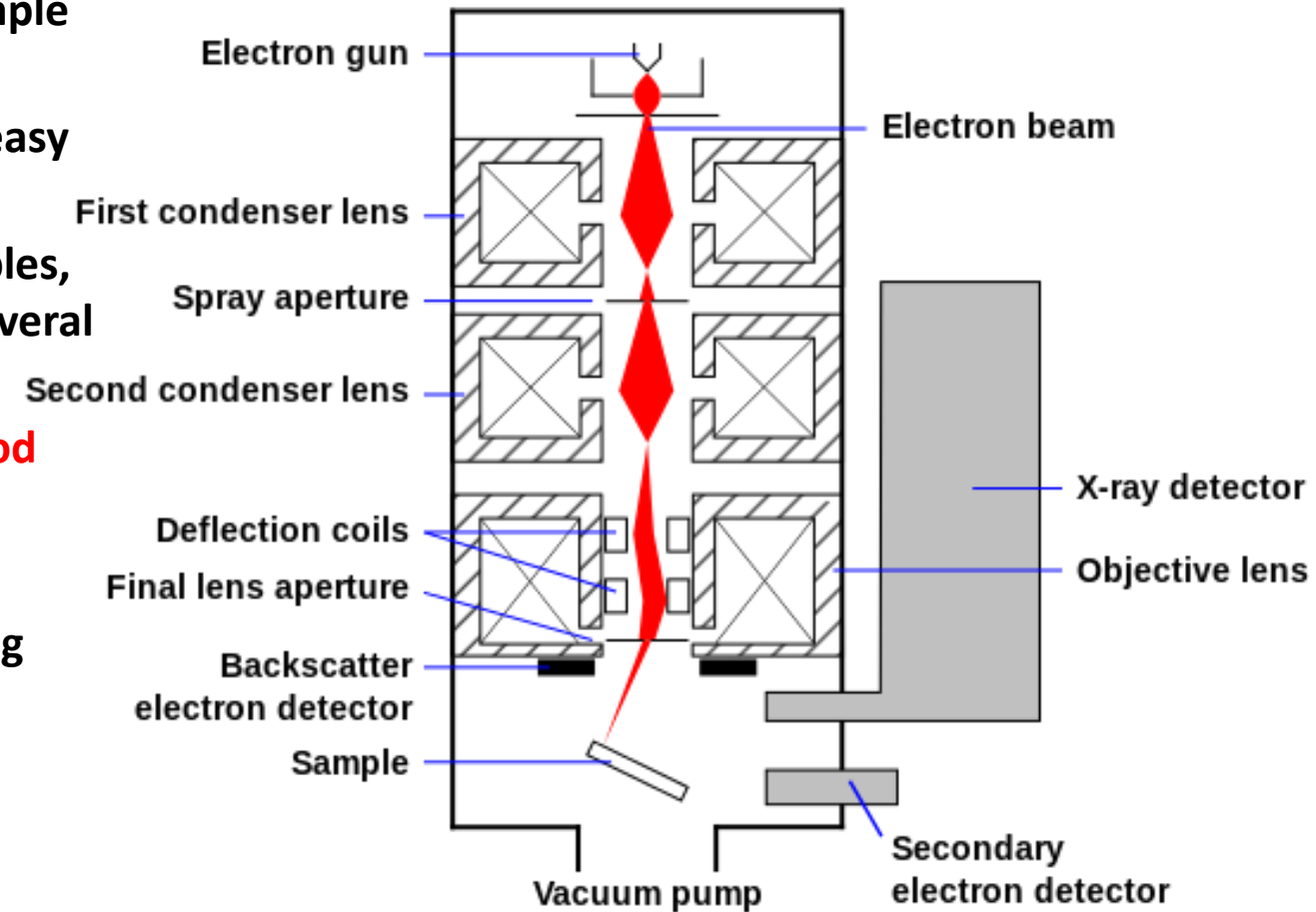
- To study defects and interfaces in detail
- The etching process can be considered as a reverse process to the growth
- The two faces of SiC behave in different ways: The Si-face is etched preferentially whereas the C-face is etched isotropically
- It is also easier to etch at dislocation and other structural defects
- The micropipes on the surface could be observed after etching in molten KOH (400 °C or above) and the number could be counted
- Different defects shapes have been observed – an evidence of screw and edge dislocations



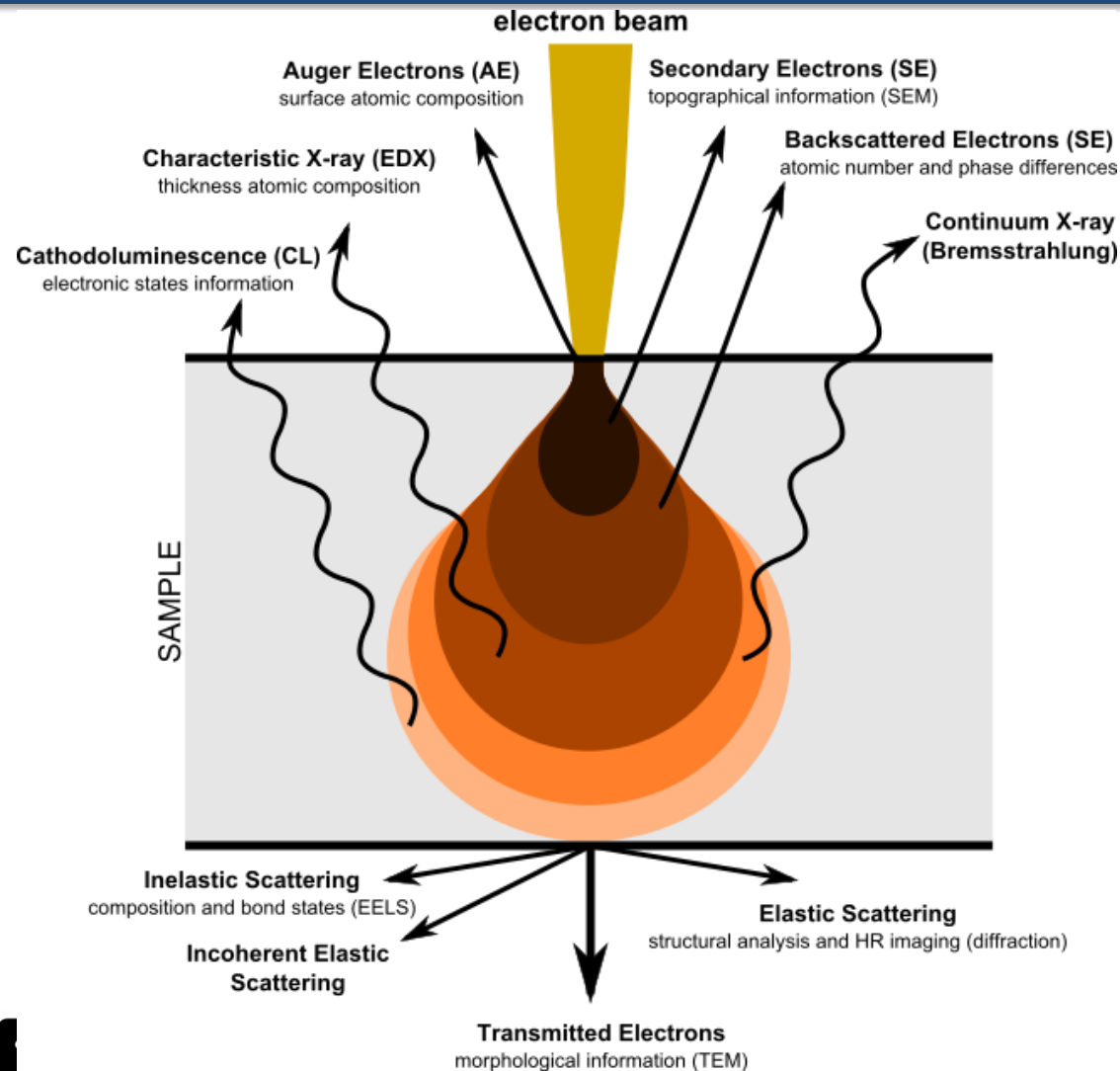
Typical micrographs of etching pattern on the C face (a) and the Si face (b) of 4H-SiC. The sample was etched by KOH at 470 °C for 20 min.

Scanning Electron Microscopy (SEM)

- SEM uses a beam of electron to scan the surface of a sample and build a three-dimensional image
- SEM is nondestructive and preparation of the sample is easy to perform
- The electron energy is typically 10–30 KeV for most samples, but for insulating samples the energy can be as low as several hundred eV
- **Advantages: high magnification, large depth of focus, good resolution**
- Two imaging modes:
 - **Primary electron imaging:** can yield images containing both topological and compositional information
 - **Secondary electron imaging:** ensures high resolution images (**Most common mode of SEM**)

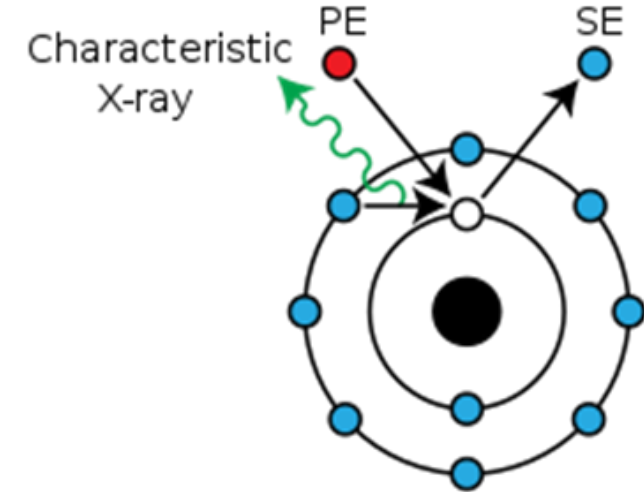
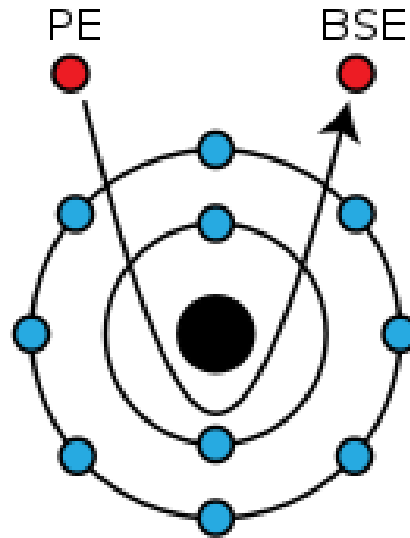
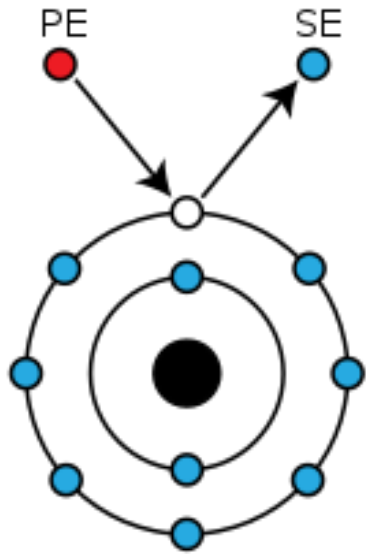


Details about SEM -1



When the primary electron beam interacts with the sample, the electrons lose energy by repeated **random scattering** and **absorption** within a teardrop-shaped volume of the specimen known as the **interaction volume**, which extends from less than **100 nm** to approximately **5 μm** into the surface. The size of the interaction volume depends on the electron's landing energy, the atomic number of the specimen and the specimen's density. The energy exchange between the electron beam and the sample results in the reflection of high-energy electrons by elastic scattering, emission of secondary electrons by **inelastic scattering** and the emission of **electromagnetic radiation**, each of which can be detected by specialized detectors.

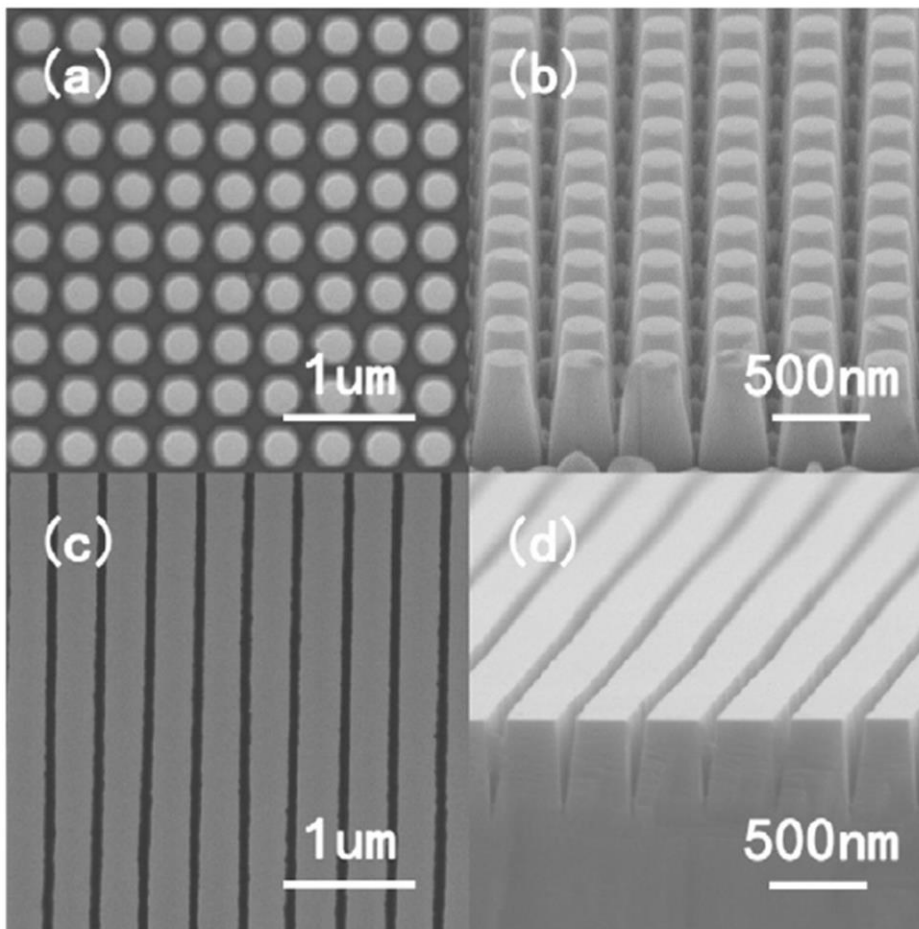
Details about SEM 2



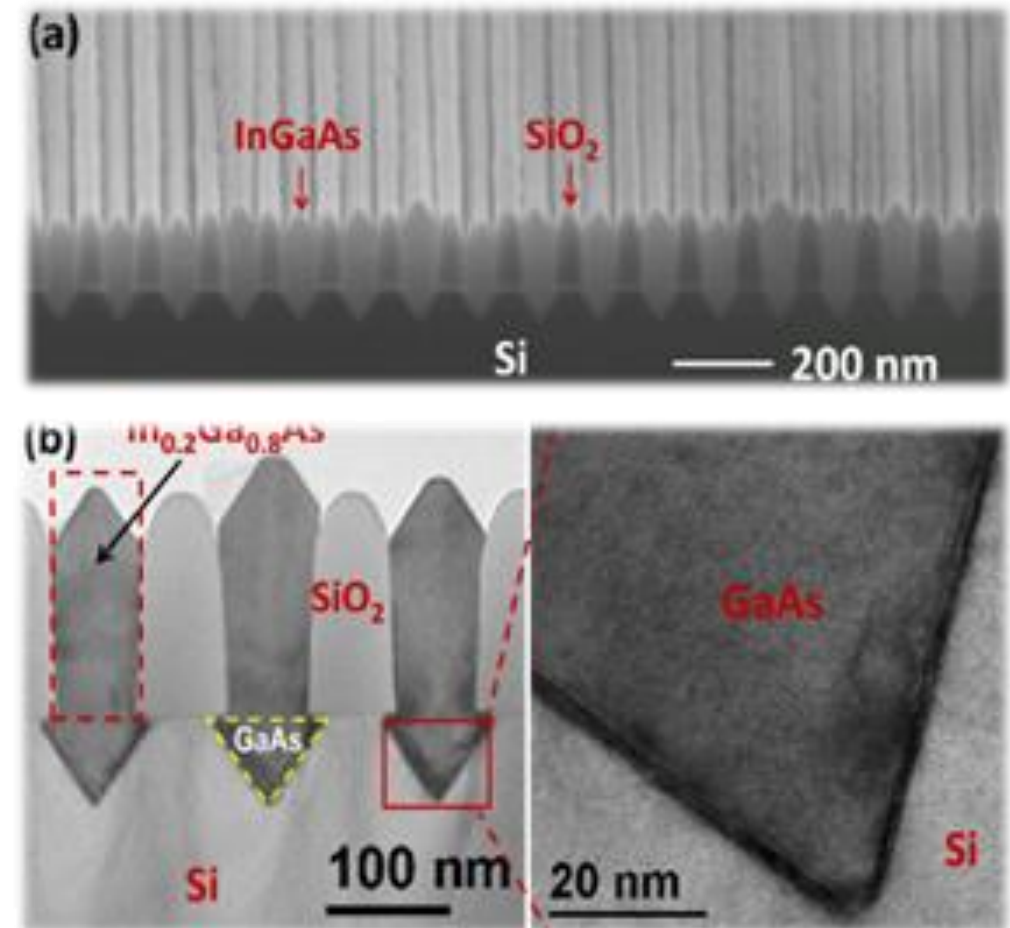
Mechanisms of emission of (a) **secondary electrons**, (b) **backscattered electrons**, and (c) **characteristic X-rays** from atoms of the sample

Details about SEM -3

Top view ((a) and (c)) and side view ((b) and (d)) SEM images of GaN nanostructures



SEM images of GaAs grown in V-groove Si substrate



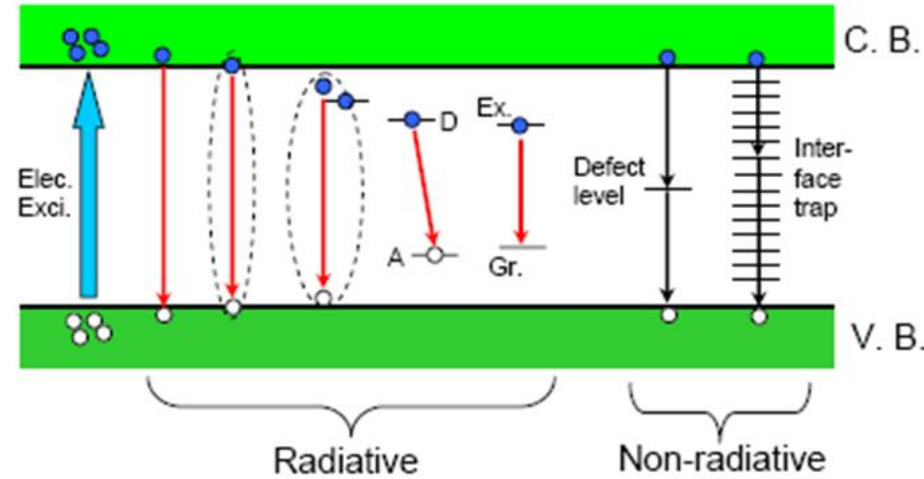
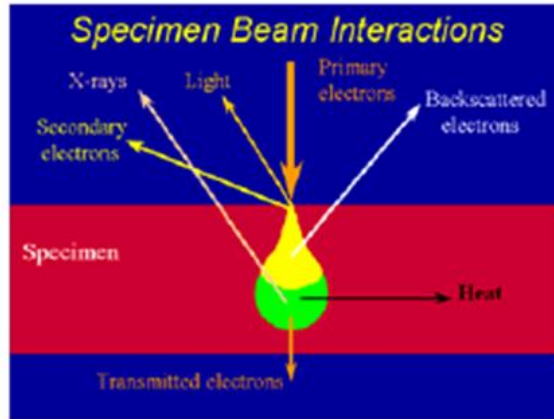


Cathodoluminescence (CL)

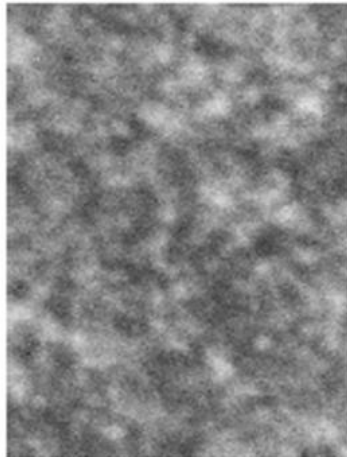
- A measurement technique to investigate the optical properties of semiconductors
- A **focused scanning electron beam** is scanned over the sample and excitation of carriers results in **luminescence**
- CL can evaluate the following:
 - ✓ **Space distribution of luminescence** and therefore the space distribution of **defects** or **impurities**
 - ✓ The influence of **mechanical defects** on the distribution of luminescence centers
 - ✓ **Electronic band structure** of the material
 - ✓ Micro-characterization of semiconductor devices



Cathodoluminescence mapping for defect characterization

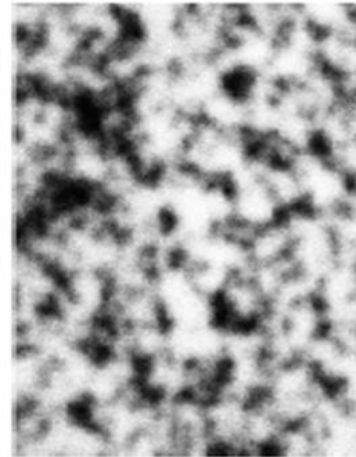


X10.25K



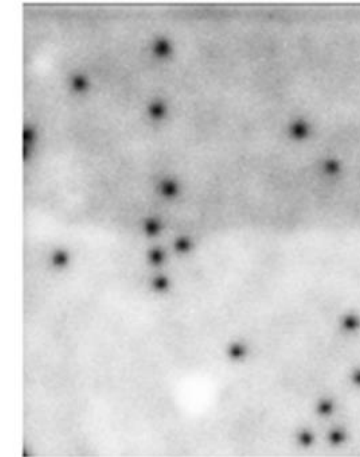
Nitronex GaN on silicon

X10.0K

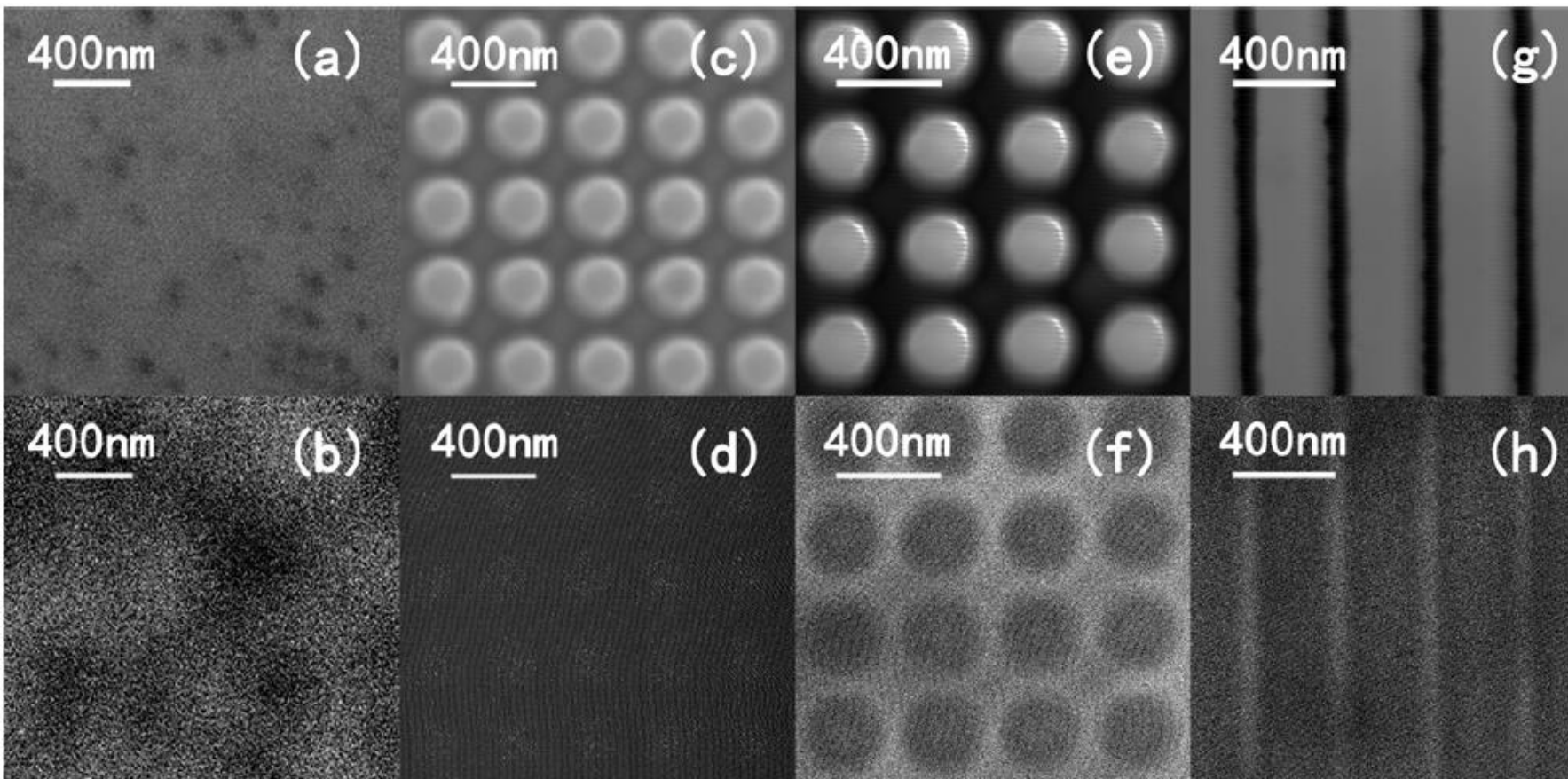


GaN on sapphire (SUNY)

X10.0K



Cree HVPE GaN substrate



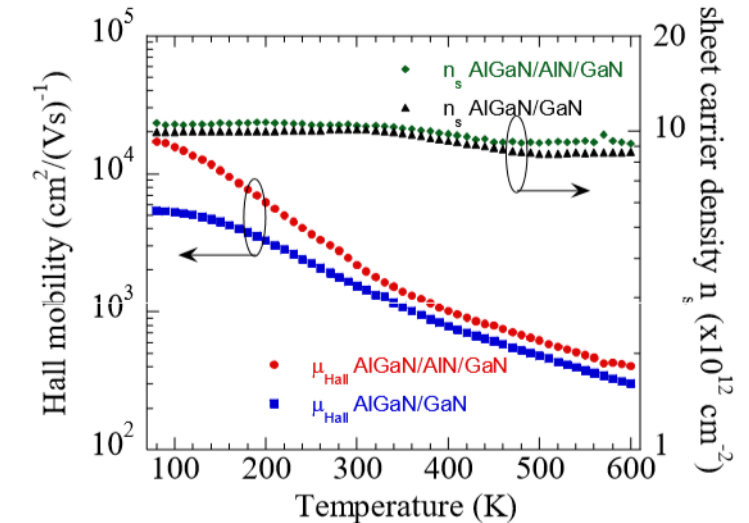
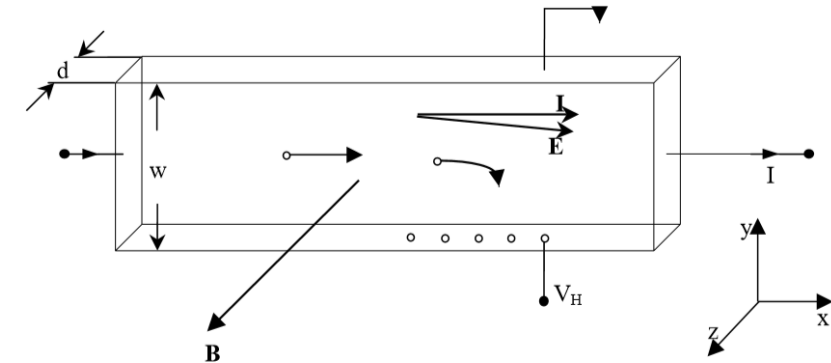
- The SEM images of GaN films and CL mappings (b) of GaN films when the acceleration voltage is 10 kV.
- The SEM images (c) and CL mappings (d) of GaN nanorods when it is 2 kV.
- The SEM images (e) and CL mappings (f) of GaN nanorods when it is 10 kV.
- The SEM images (g) and CL mappings (h) of nano-gratings when it is 10 kV. (a)

Zhuang, Zhe, et al. "Large-scale fabrication and luminescence properties of GaN nanostructures by a soft UV-curing nanoimprint lithography." *Nanotechnology* 24.40 (2013): 405303.



Hall Effect measurements

- If an electrical current I flows through a semiconductor in a magnetic field B , the moving charge carriers experiences a transverse voltage which pushes them to one side of the semiconductor,
- There is a **build-up of charges** at the sides of the semiconductor and will balance the magnetic influence, leading to a measurable voltage between the two sides.
- **The transverse voltage V_H is called the Hall Effect** (E.H. Hall discovered it in 1879)
- The **hall voltage has a deferent polarity** in the case of positive and negative charge carriers
- Hall effect can measure **carrier mobility** and **carrier density** in semiconductors
- Hall effect measurement can be performed at varied temperatures



Hall effect -2

- Consider a p-type semiconductor bar with
- a magnetic field B_z acting in the z direction, and an electric field E_x acting in the x direction.
- A hole current is established in the x-direction.
- The **Lorentz force** due to the magnetic field on these moving holes is given by :

$$F_y = - qv_x B_z$$

and is acting in the -ve y-direction.

- transport of holes by this force results in a pile up of holes in the boundary in the -ve y-direction, thus giving rise to an electric field E_y .

- The establishment of this transverse field E_y is called the **Hall Effect**, and the resulting voltage is called the **Hall Voltage**.
- **E_y balances the Lorentz force** due to the magnetic field, an equilibrium in the forces acting along the y-direction is established when :

$$qE_y = qv_x B_z$$

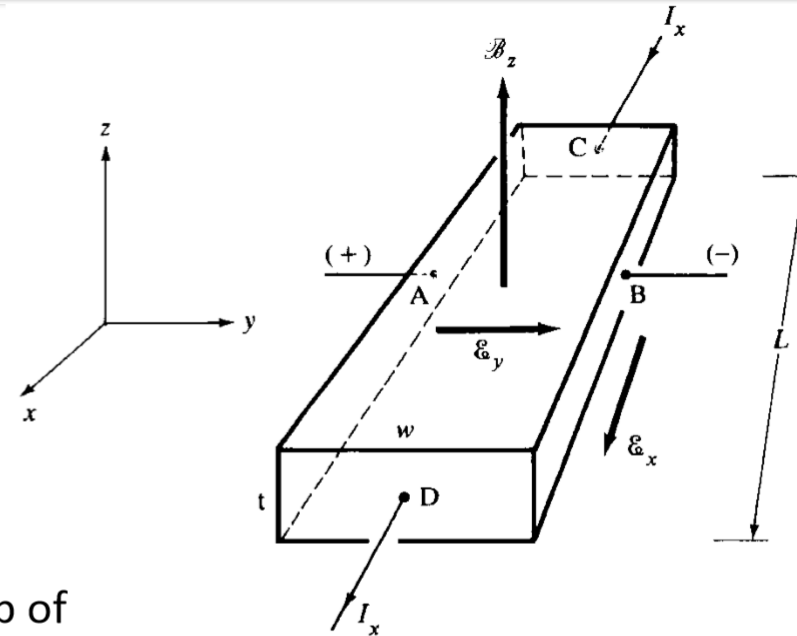


Fig. 3-25 The Hall effect.

Hall effect -3

Therefore : $E_y = v_x B_z$ But $v_x = \frac{J_x}{qp_o}$
 we have $E_y = \frac{J_x}{qp_o} B_z = R_H J_x B_z$
 where $R_H \equiv \frac{1}{qp_o}$ is defined as the Hall Coefficient.

Alternatively : $p_o = \frac{1}{qR_H} = \frac{1}{q} \frac{J_x B_z}{E_y} = \frac{1}{q} \frac{I_x}{wt} \frac{B_z w}{V_{AB}} = \frac{I_x B_z}{qtV_{AB}}$ where all the quantities on the RHS can be measured.

Hence

- The hole concentration (p_o) can be measured once I_x , V_{AB} , and B_z are determined.
- Since the conductivity $\sigma = q\mu_p p_o$

Hence:

$$\mu_p = \frac{\sigma}{qp_o} = \frac{1}{\rho} \frac{1}{q} qR_H = \frac{R_H}{\rho}$$

i.e. The mobility is simply the ratio of the Hall Coefficient and the resistivity.

The resistivity can be easily determined by :

V_{CD} and I_x can be measured and all the dimensions are known.

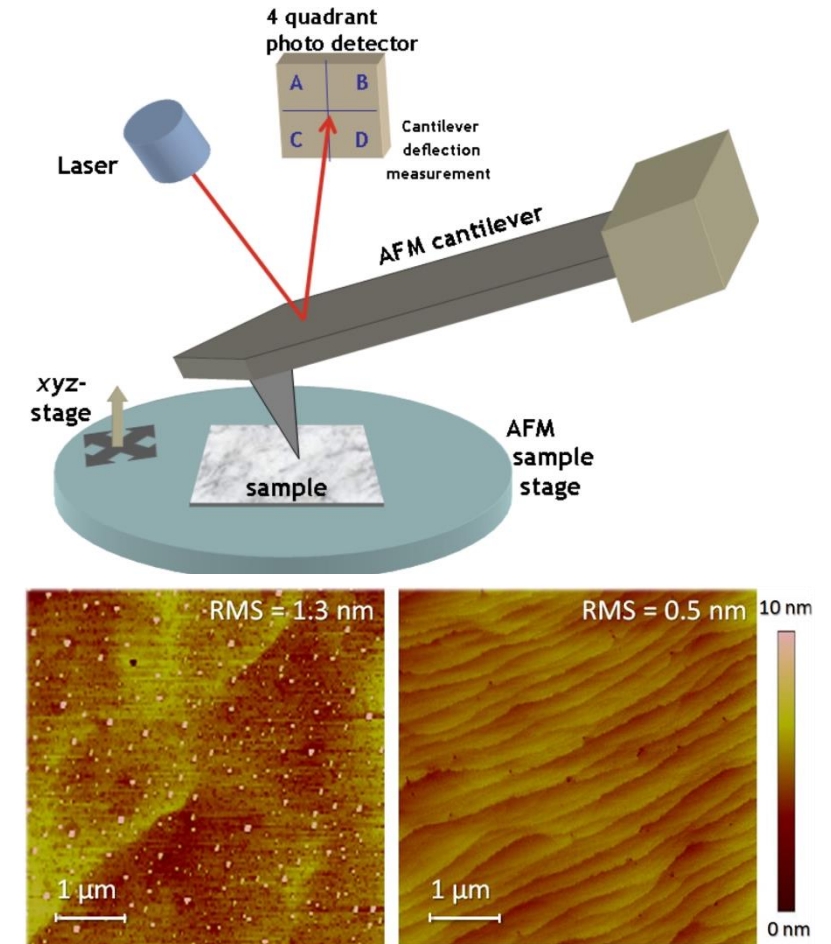
$$\rho = \frac{wt}{L} R = \frac{wt}{L} \frac{V_{CD}}{I_x}$$

- For an n-type material, both V_H and R_H are negative quantities => it is possible to **determine the type of the material by inferring from the sign of the Hall Coefficient.**



Atomic Force Microscopy (AFM)

- AFM is a **high-resolution surface sensitive** measurement technique
- AFM consists a **cantilever with a sharp tip** to probe the surface. The tip is of few nanometers in radius of curvature. While approaching the surface, the **tip is being deflected by forces** as chemical bonding, Van der Waals force, electrostatic force or magnetic force. Deflected **tip pushes up the cantilever**. by measuring the changes in the tip movement by **laser deflection**, the image of the surface can be obtained.



AFM images for p-GaN surface of an LED wafer and n-GaN surface after regrowth.



TEM – history

- 1925 Louis de Broglie first theorized that the electron had wave-like characteristics, with a wavelength substantially less than visible light. (He received the Nobel Prize in 1929) ($\lambda = h/p = h/(mv)$)
- 1927 Davisson and Germer in USA, Thompson in UK independently carried out their classical electron diffraction experiments which demonstrated the wave nature of electrons. (Davisson and Thompson received the Nobel Prize in 1937) (electron diffraction)
- 1932 Knoll and Ruska designed and built the first TEM in the world. (Ruska received the Nobel Prize in 1986.)
- 1933 one year later, the resolution limit of light microscope (0.2 μ m resolution) was surpassed by TEM





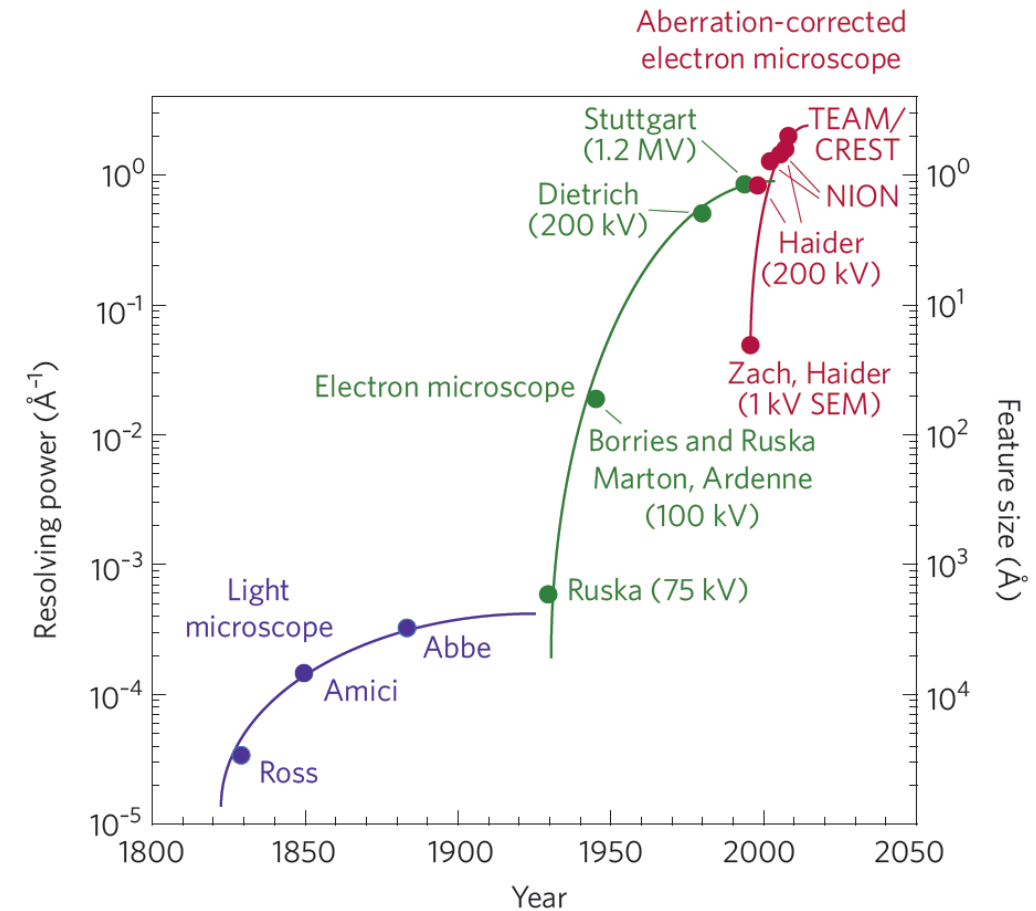
TEM – history -2

- 1936 The first commercial TEM, the Metropolitan-Vickers EM1, was built in the UK, but not very successful.
- 1939 regular production was really started by Siemens and Halske in Germany.
- 1945 after World War II, many other companies started to produce TEMs. [Hitachi, JEOL (Japan); Philips (Holland); RCA(Radio Corporation of America) (USA)]
- 1949 Heidenreich first thinned metal foils to electron transparency which is very important for the materials scientists to start use TEM as a powerful tool for research.
- 1950' s Hirsch' s group in Cambridge (UK) developed the theory of electron diffraction contrast with which we can now identify all known line and planar crystal defects in TEM images.



TEM – history-3

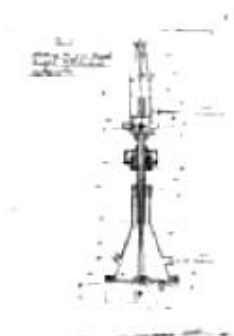
- 1960' s High voltage electron microscopes (HVEMs) was developed with accelerating voltage between 1 MV and 3 MV.
- 1970' s many commercial TEMs have been capable of resolving individual columns of atoms in crystals, creating the field of "high resolution transmission electron microscopy " (HRTEM).
- 1980' s intermediate voltage electron microscopes (IVEMs) were introduced which operate at 300 to 400 kV with very high resolution, close to the 1MV machine.
- 1980' s the first high resolution HVEM (1MV) was developed and into 1990' s more high resolution HVEM were built (1.25 MV) mostly in Japan.
- 21st century the Cs-corrected TEM and mono-chromatic TEM were developed which bring the resolution into sub-Å and sub-eV ranges.





Key Stages Development of TEM

1930'
电镜诞生



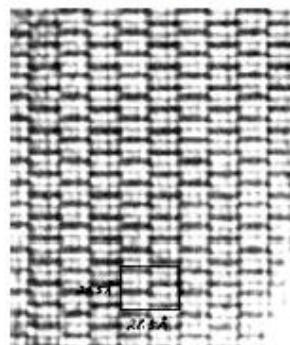
E. Ruska

1950-70'
衍射衬度



P.B. Hirsch

1970-90'
相位衬度

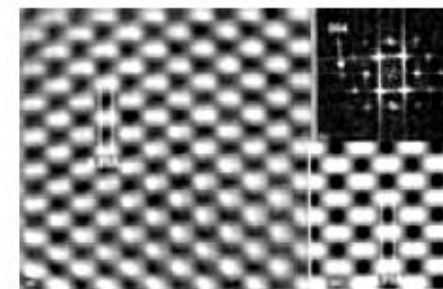


J.M. Cowley

1980-2000'
分析电子显微学

EDS
EELS
END
CBED
REM
STEM

2000-
亚埃尺度



A. Howie



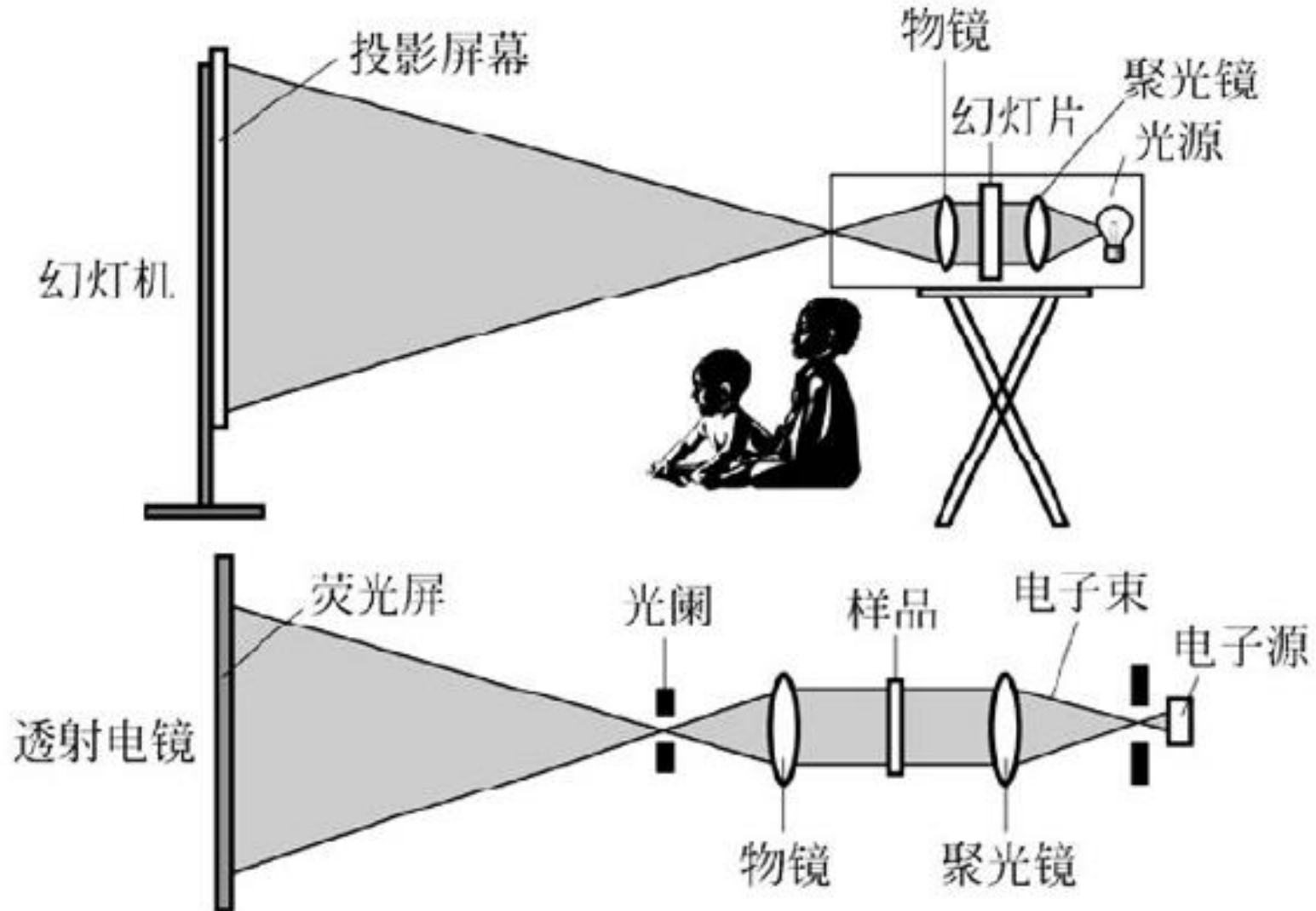
M.J. Whelan



S. Iijima

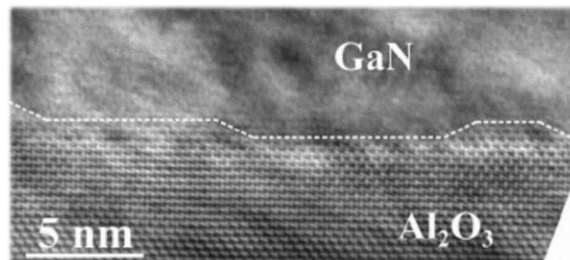


TEM is like a projector

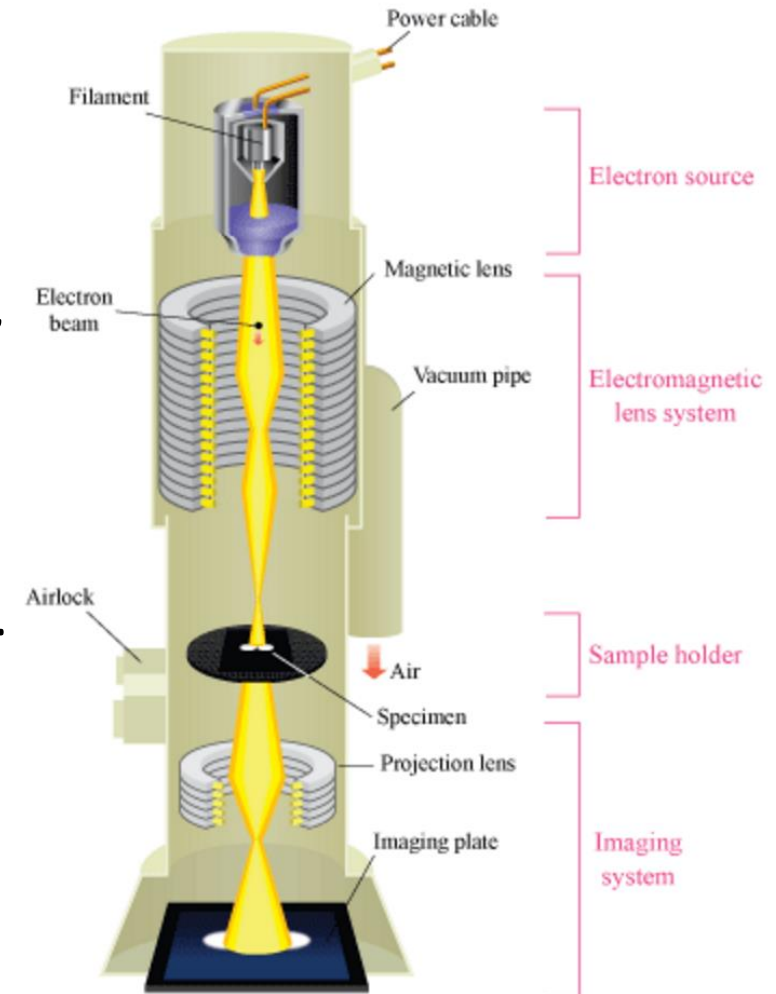


TEM equipment structure

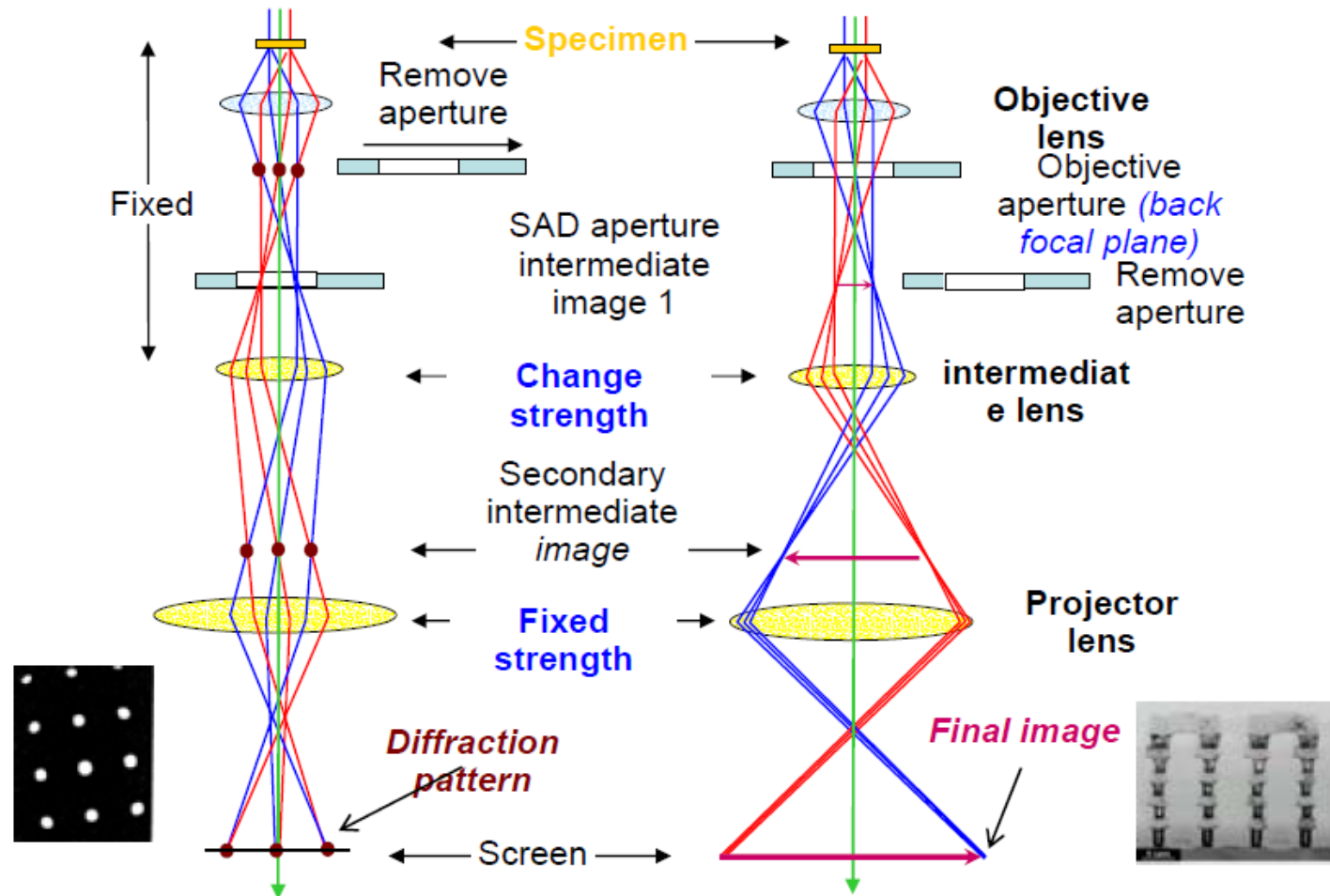
- **Transmission Electron Microscope (TEM)** utilizes energetic electrons to provide morphologic, compositional and crystallographic information on samples.
- At a maximum **potential magnification of 1 nanometer**, TEMs are the most powerful microscopes. TEMs produce high-resolution, two-dimensional images, allowing for a wide range of educational, science and industry applications.
- During transmission, **the speed of electrons directly correlates to electron wavelength**; the faster electrons move, the shorter wavelength and the greater the quality and detail of the image.
- To obtain a TEM analysis, samples need to have certain properties. They need to **be sliced thin enough for electrons to pass through**, a property known as electron transparency.



TEM image of an thin GaN/Al₂O₃ interface



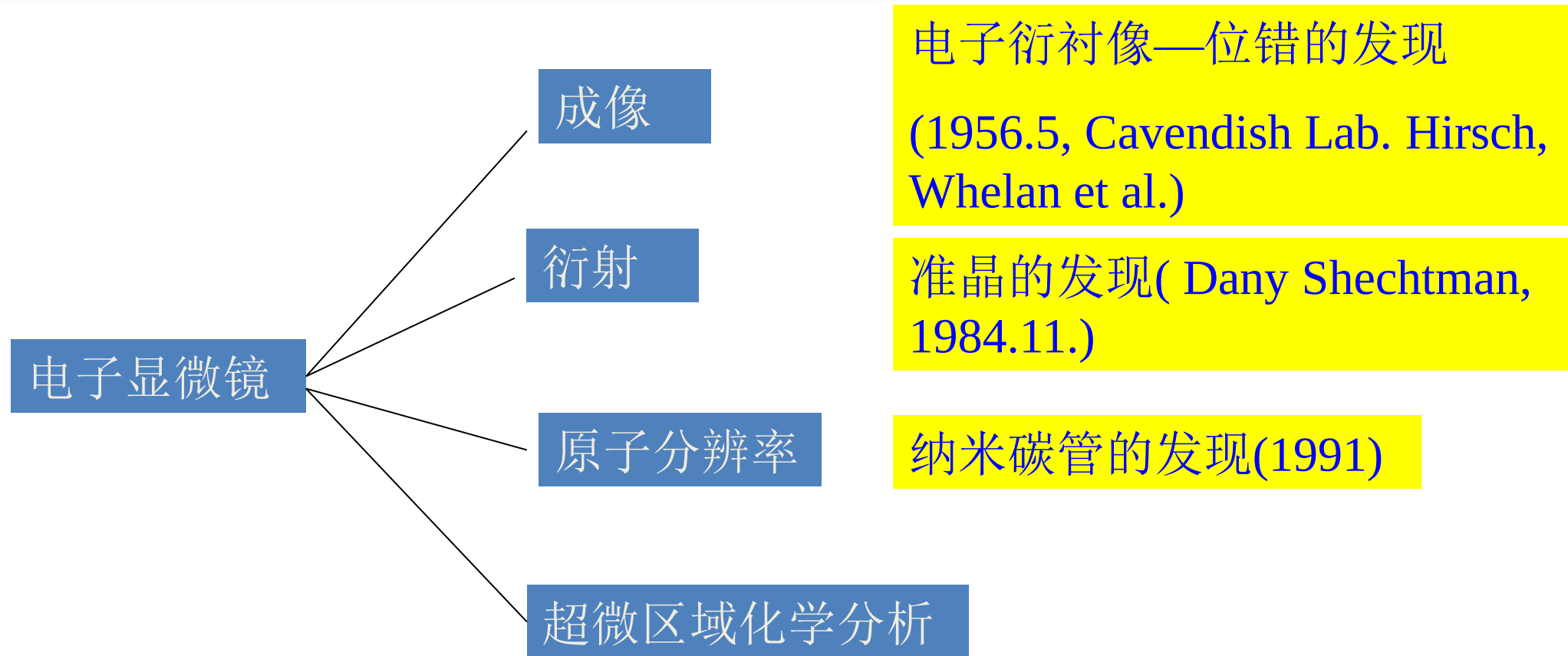
Diffraction mode & Image mode



There are two basic modes of TEM operation, namely the **bright-field mode**, where the (000) transmitted beam contributes to the image, and the **dark-field imaging mode**, in which the (000) beam is excluded. The size of the objective aperture in bright-field mode directly determines the information to be emphasized in the final image. When the size is chosen so as to exclude the diffracted beams, one has the configuration normally used for low-resolution defect studies, so-called **diffraction contrast**. In this case, a crystalline specimen is oriented to excite a particular diffracted beam, or a systematic row of reflections, and the image is sensitive to the differences in specimen thickness, distortion of crystal lattices due to defects, strain and bending.



Future development for (HR)TEM



当前各类学术期刊发表的自然科学论文, 超过50%的研究工作都使用到电子显微技术!!



TEM can do many things

- **Atomic structure**

- Diffraction
(lattice parameters,
orientations) **$\sim 1\text{pm}$**
- High resolution TEM
 $\sim 0.2\text{nm}$

- **Microstructure
& Defects**

- Bright field image
- Dark field imaging
 $\sim \mu\text{m} - \text{nm}$

Elastic scattering

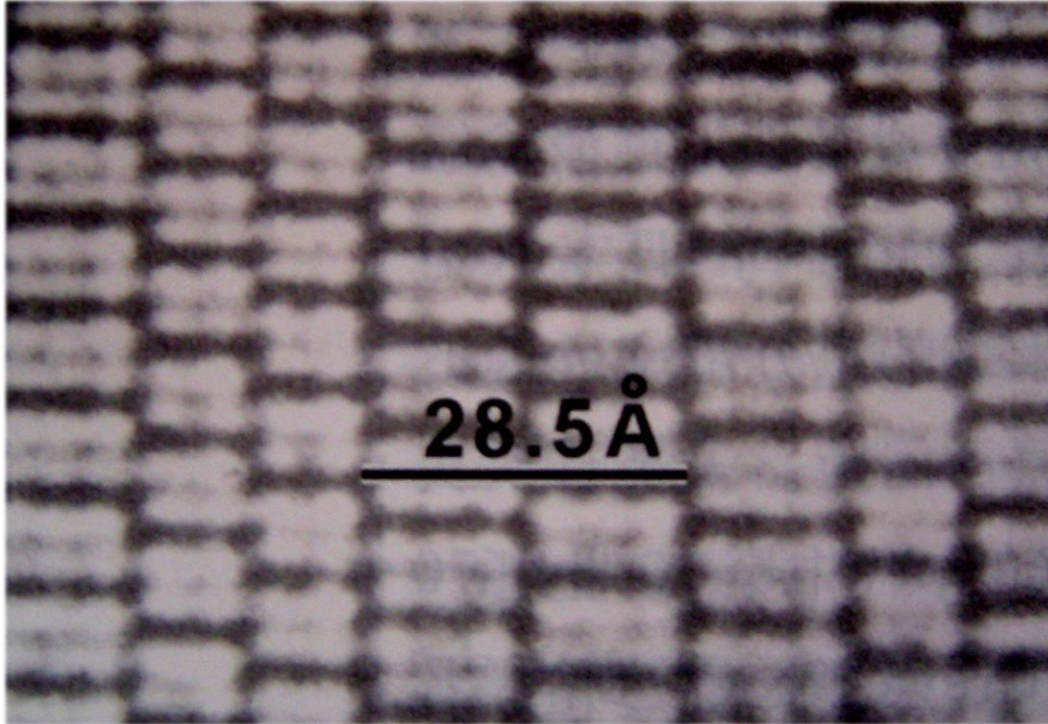
- **Chemistry**

- Energy dispersive X-ray microanalysis
EDX, **$\sim 1\text{nm}$**
- Electron Energy Loss Spectroscopy
EELS

Inelastic scattering

**In-situ
Cooling/Heating**

First TEM image



历史上首张二维原子分辨率电子显微像, 点分辨率大约 0.35 nm, 由饭岛(Sumio Iijima)沿 $\text{Ti}_2\text{Nb}_{10}\text{O}_{29}$ 化合物的b轴拍摄, 发表于J. Appl. Phys., 42 (1971) 5891.

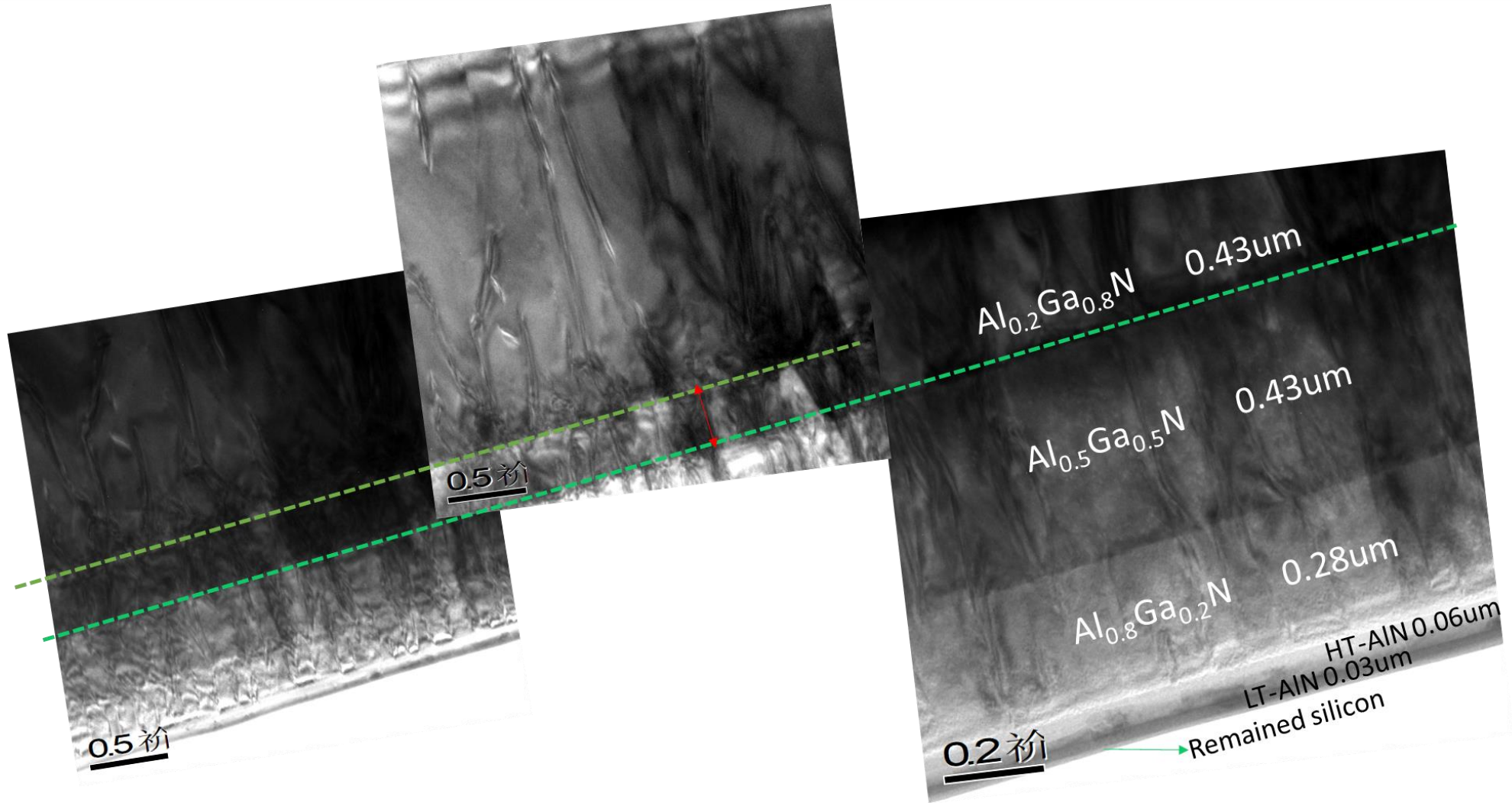
图中灰色的点对应电子束方向投影的金属原子柱, 黑线处对应着距离太近尚不能分辨的原子

日本JEM-100B (100kV)

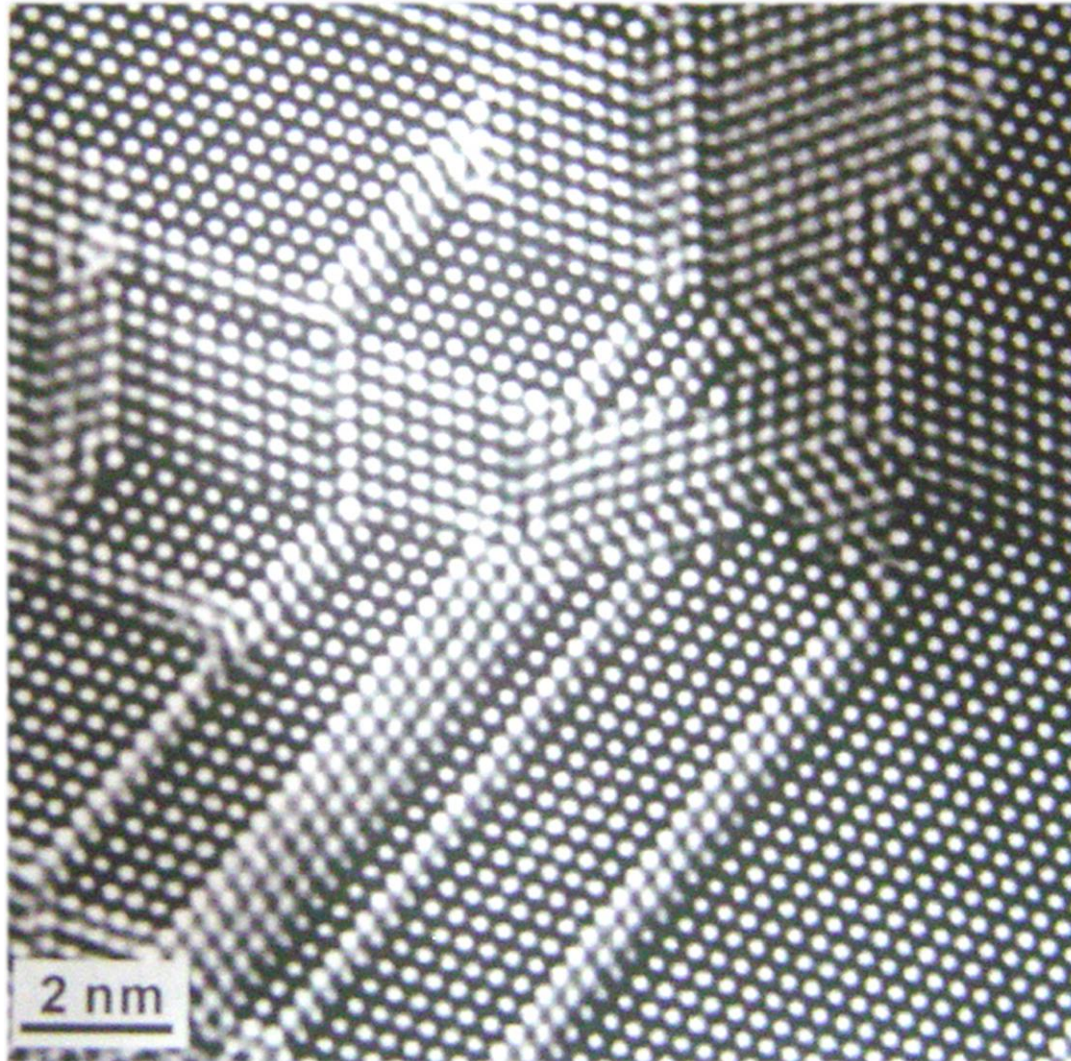
饭岛和考利(John Cowley)-----获美国物理学会
1972年度奖
考利获1979年美国电镜学会物理科学杰出科学家大奖



TEM images of GaN on Si

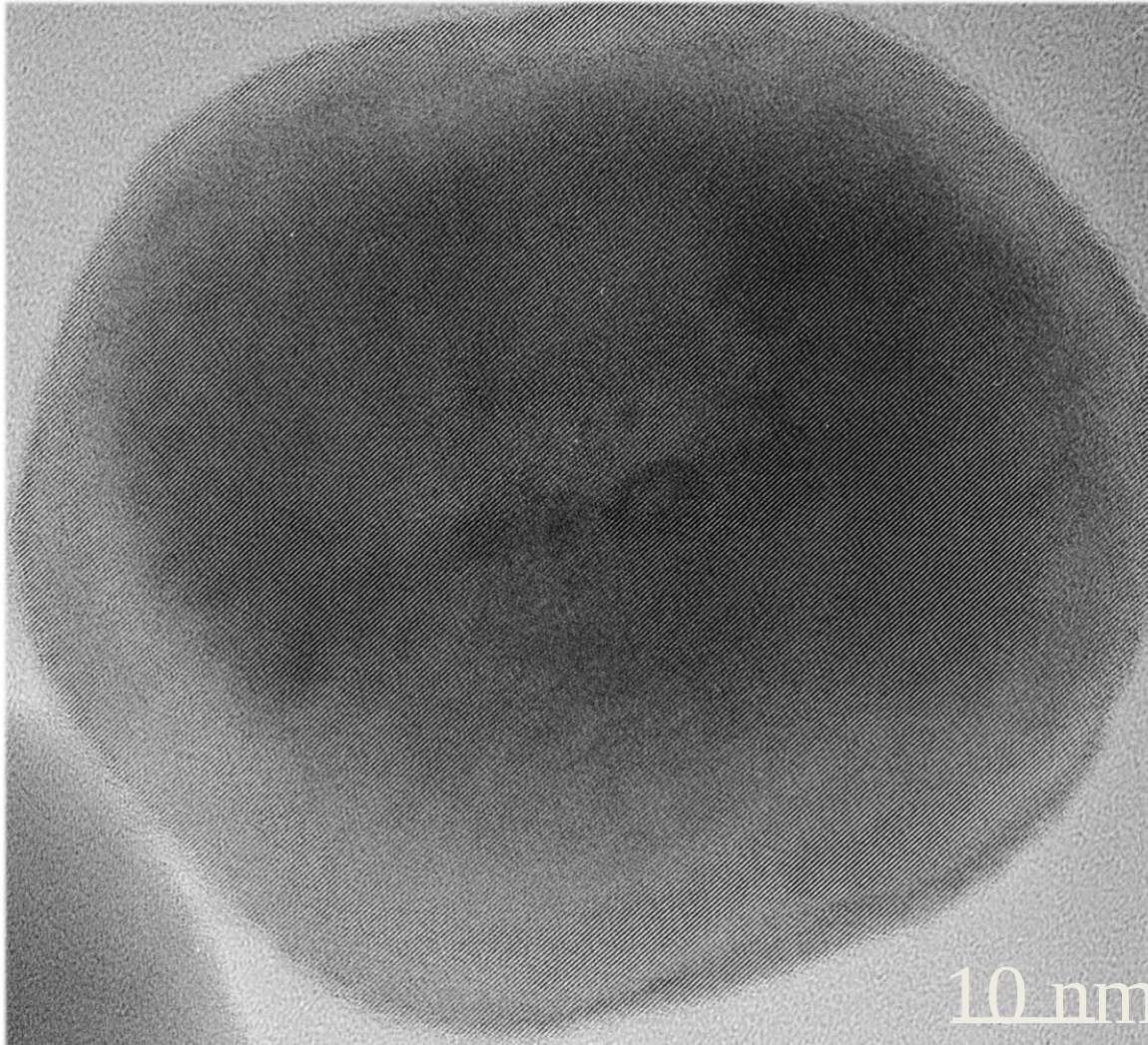


TEM images



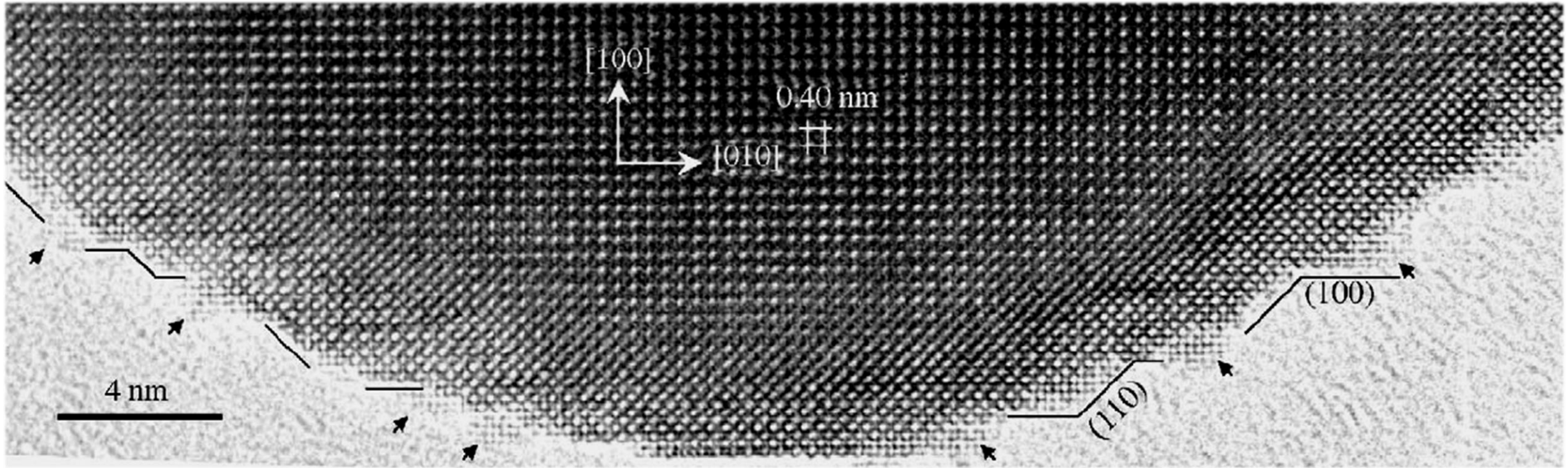
超高压电镜拍摄的[110] 带轴
Si 原子显微像, 白色像点对应
沿电子束方向投影的Si 原子,
可见到多重孪晶

TEM images



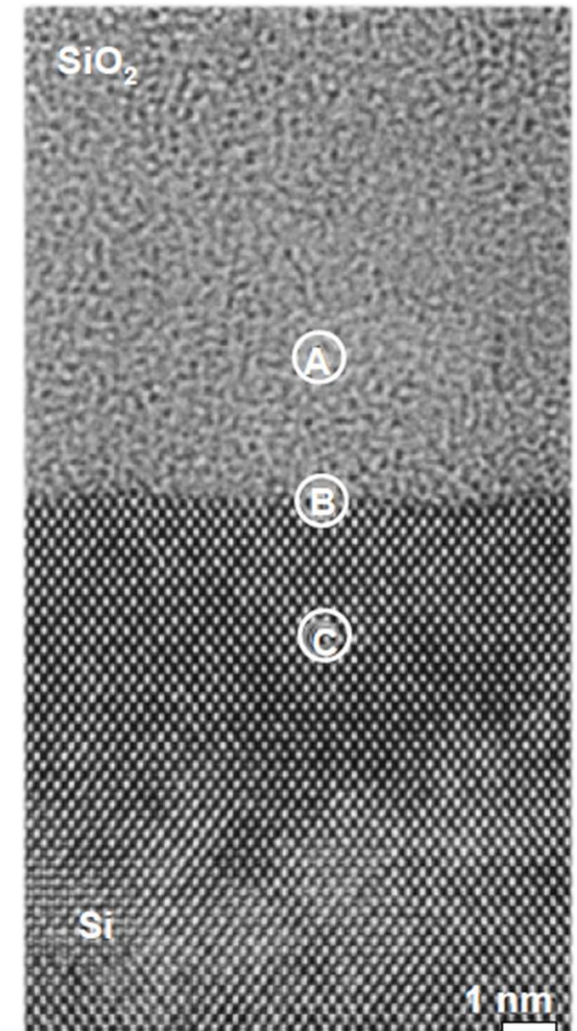
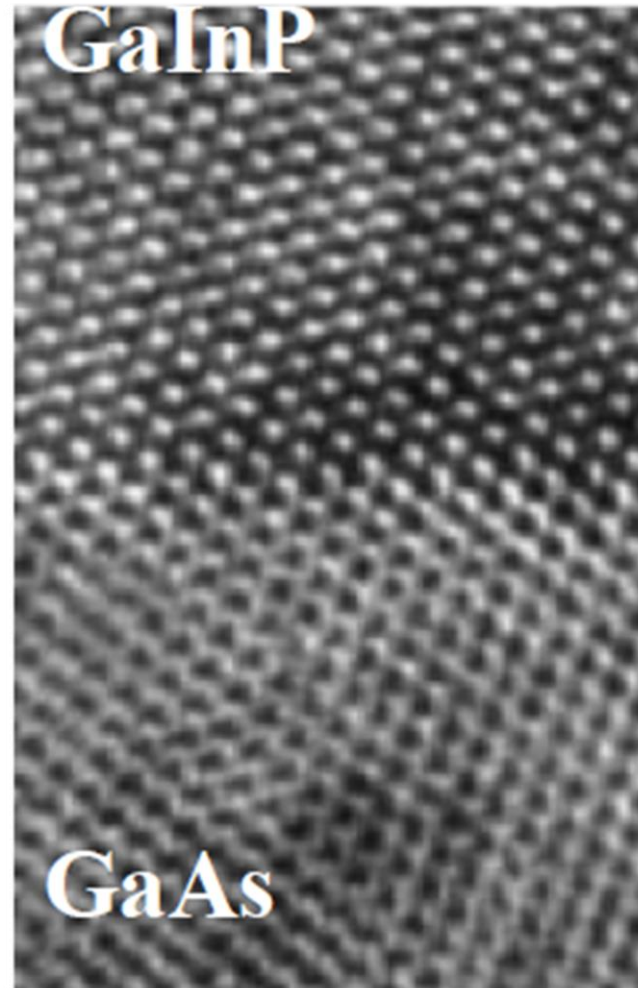
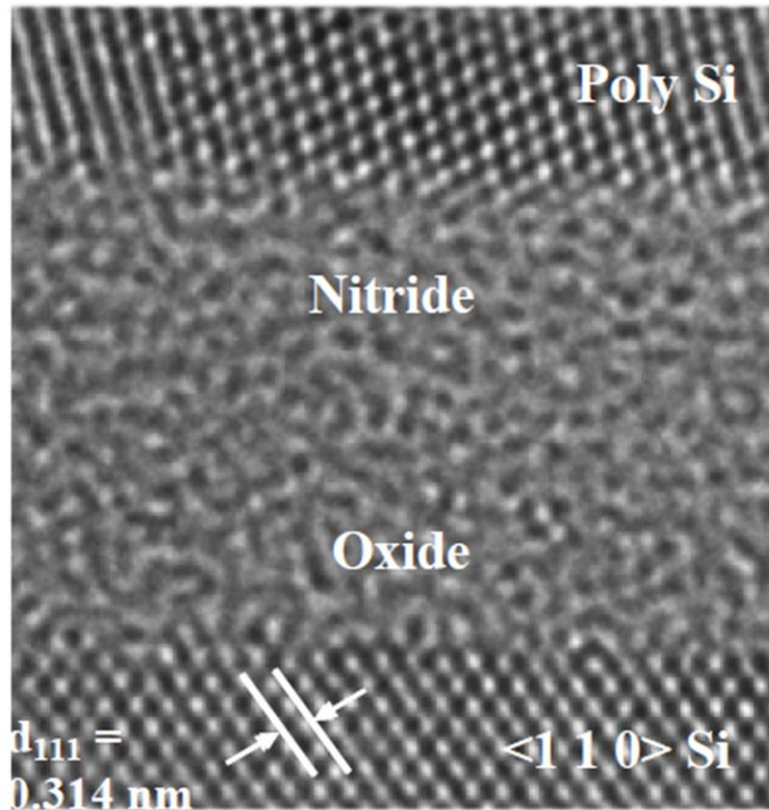
lattice image of a nano-crystalline
 BaTiO_3 grain with size of 75 nm

TEM images



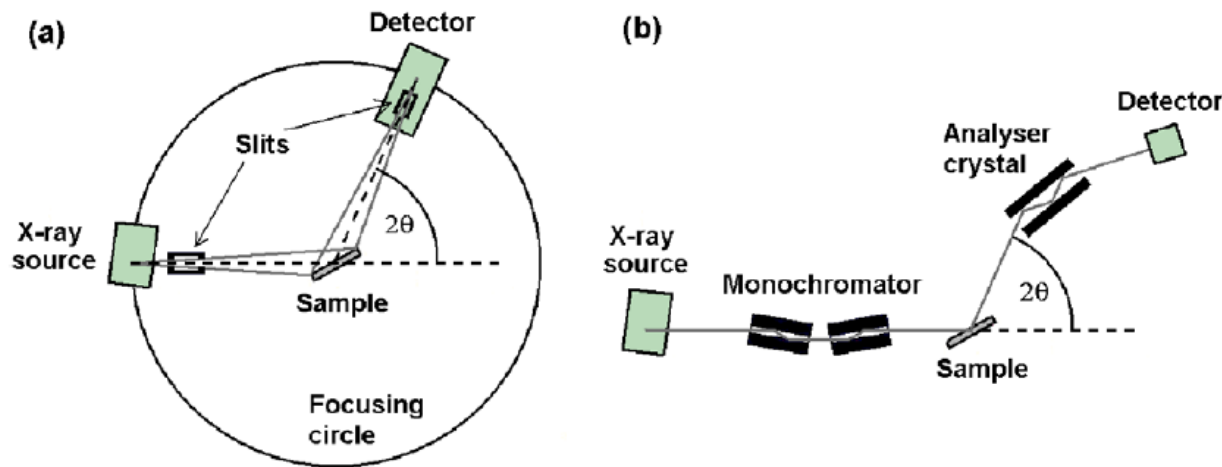
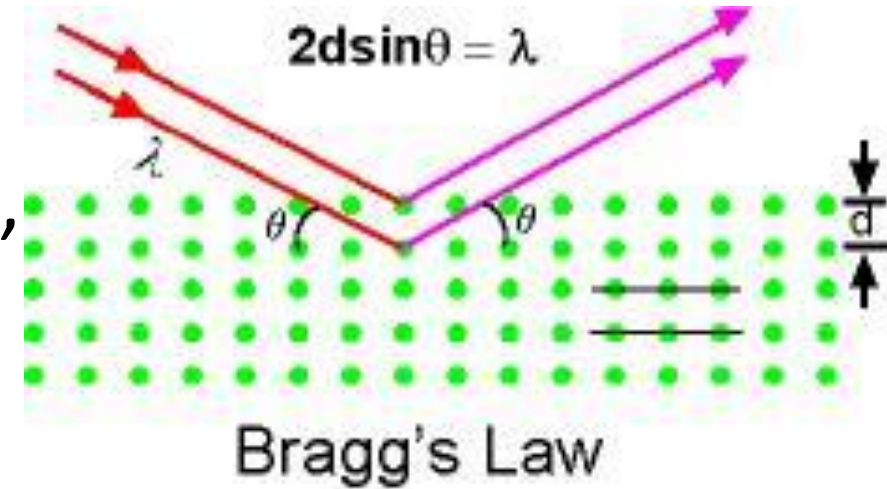
A surface profile HRTEM image of part of a BT grain with size of 80 nm

HREM Image — Interface



XRD

- XRD- x-ray diffraction
- Investigate: **crystalline material structure**, including atomic arrangement, crystallite size, and imperfections



Example geometries for (a) powder diffraction (no incident beam monochromator) and (b) high-resolution diffraction (incident beam monochromator and precision goniometer present, analyser crystal optional). Many other combinations of optics are possible

XRD curve of an AlGa_N/Ga_N HEMT

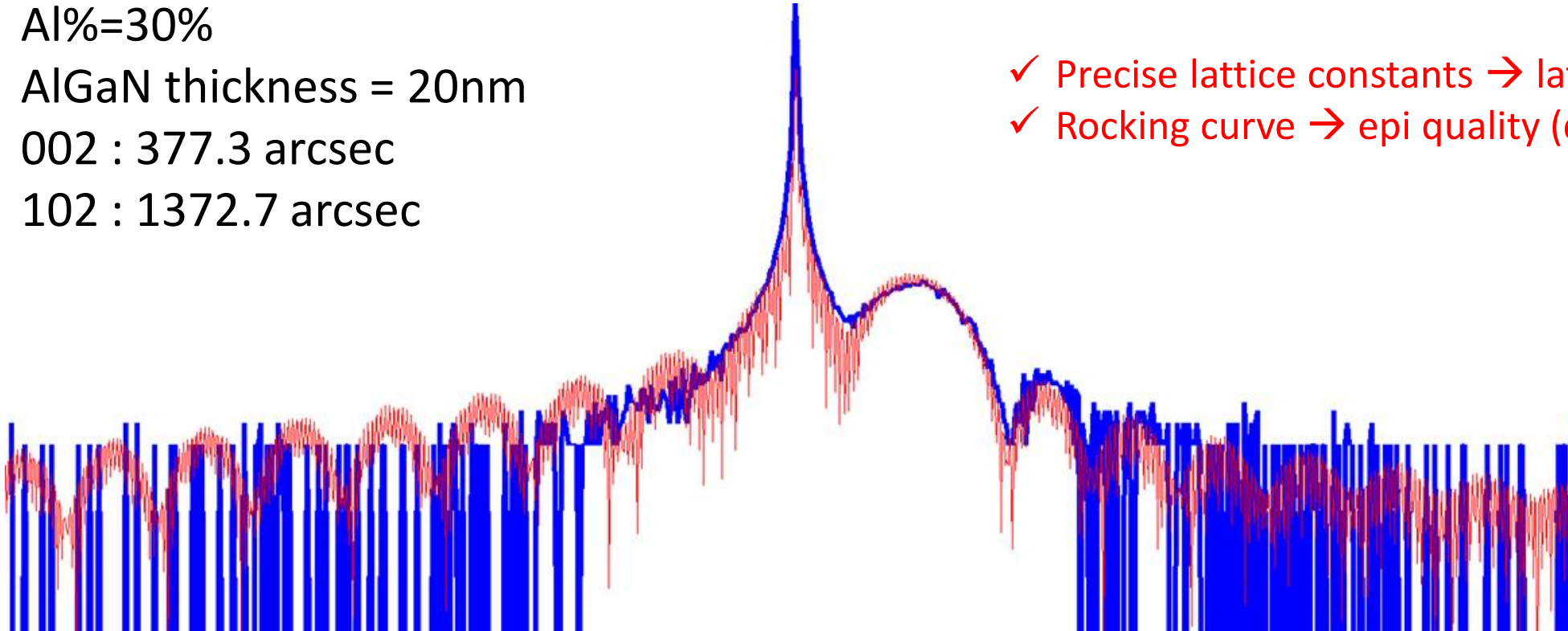
Al%=30%

AlGa_N thickness = 20nm

002 : 377.3 arcsec

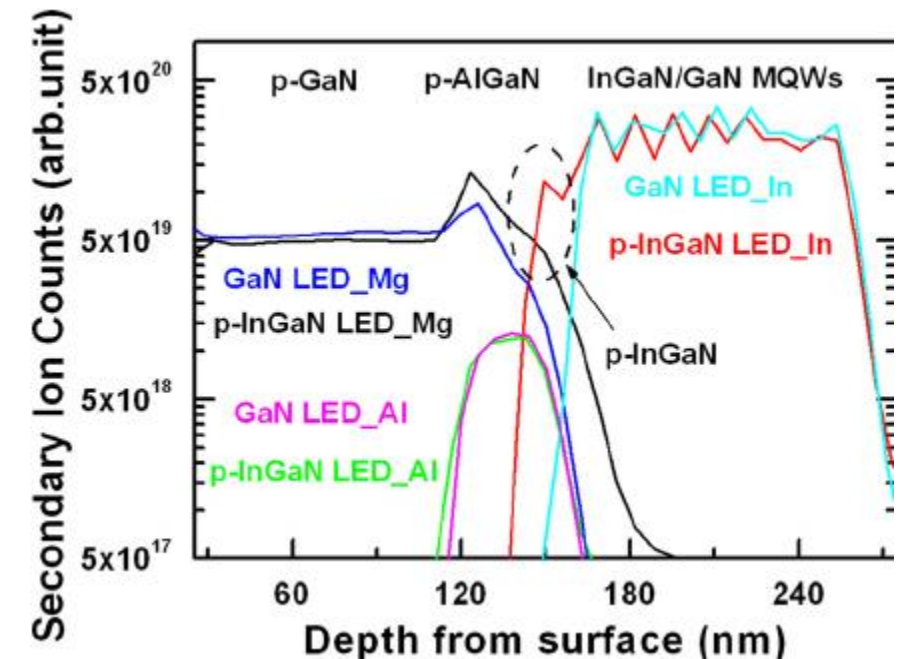
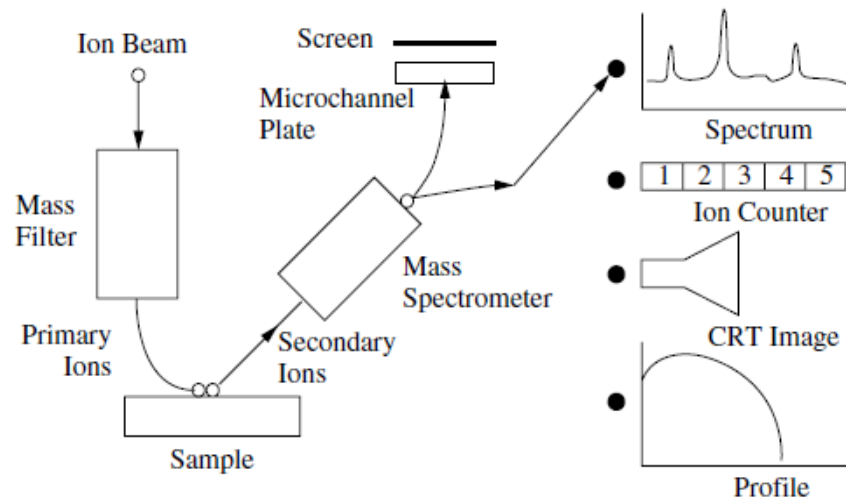
102 : 1372.7 arcsec

- ✓ Precise lattice constants → lattice mismatch
- ✓ Rocking curve → epi quality (dislocation density)



SIMS

- SIMS-Secondary ion mass spectrometry
- Element specific and is capable of **detecting all elements** as well as isotopes (同位素) and molecular species
- Lateral resolution is typically 100 μm but can be as small as 0.5 μm with depth resolution of 5 to 10 nm



SIMS results of a GaN LED

XPS

- XPS-X-ray photoelectron spectroscopy
- For identifying chemical species at the sample surface, allowing all elements except hydrogen and helium
- Surface sensitive: 0.5–5 nm
- For identification of compounds using **energy shifts** due to changes in the chemical structure of the sample atoms

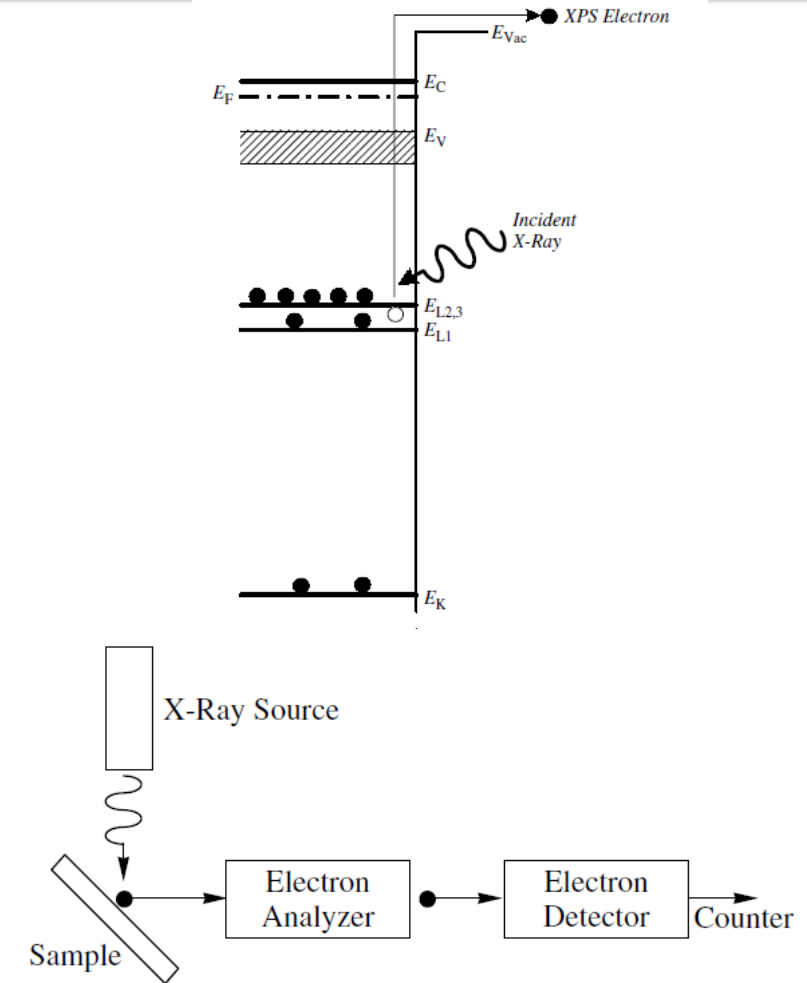


illustration of photoelectron emission

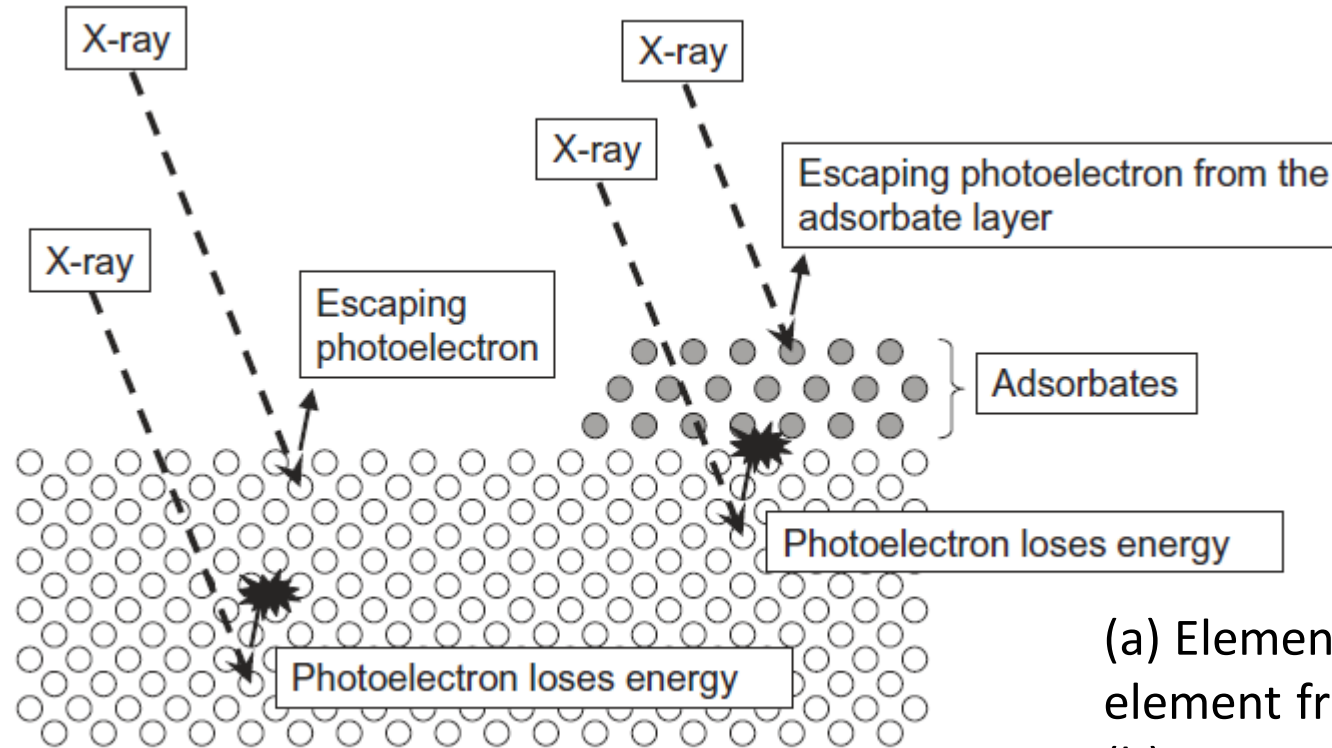


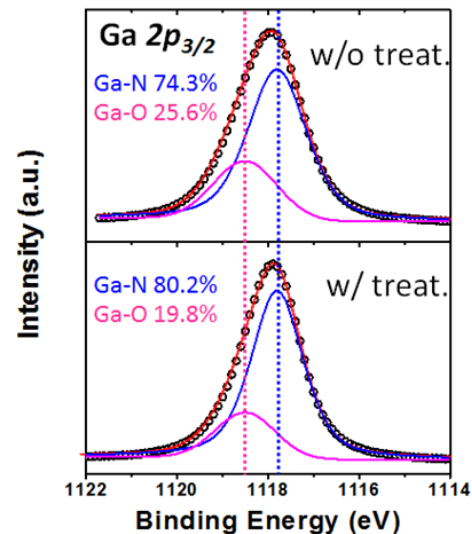
Illustration of photoelectron emission (those with a $K.E.$ of ~ 100 eV) from a solid material (represented by the hollow circles) and the surface adsorbate layer (represented by the gray circles).

- (a) Elemental identification and quantification of any element from Li to U
- (b) Sensitivity (concentrations down to 0.1 atomic %)
- (c) Surface specificity (less than 10 nm)
- (d) Ease of analysis (minimal sample preparation is required)

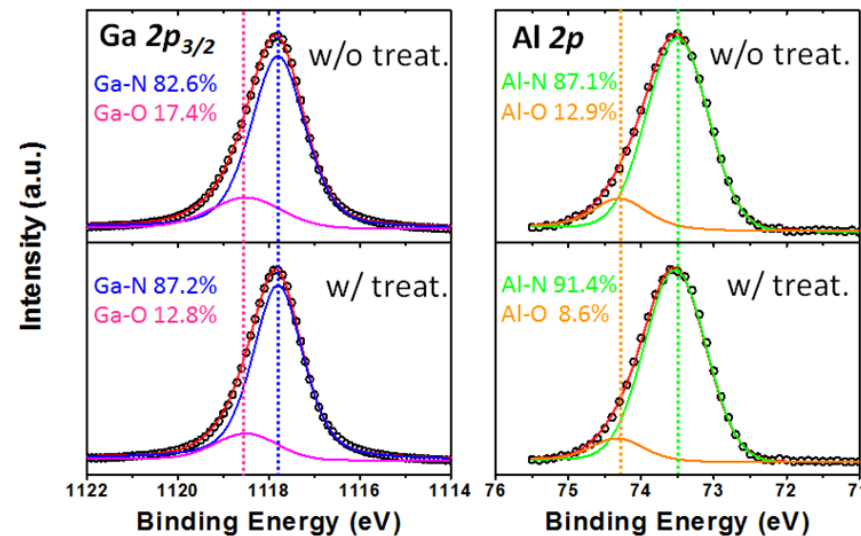
XPS

- XPS characteristics for AlGaIn/GaN HEMT surface before and after O₂ plasma & HCl treatment

(a) Exposed GaN buffer surface

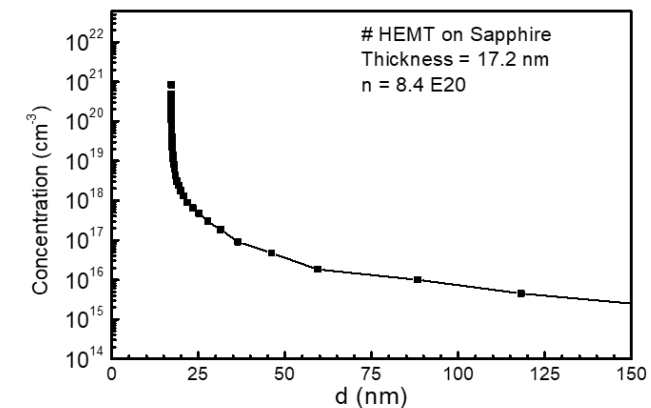
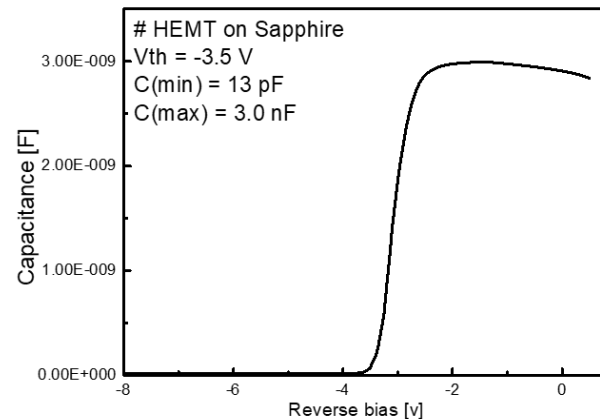
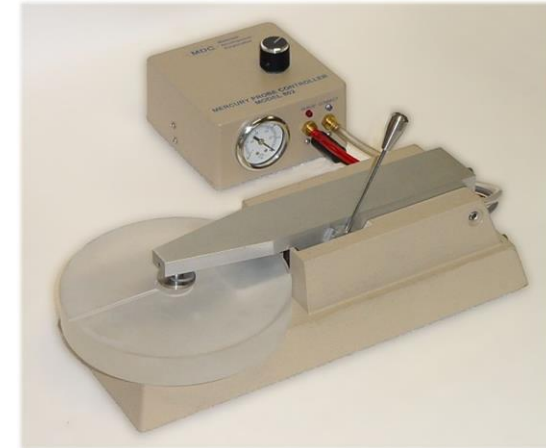


(b) AlGaIn barrier surface



Mercury Probe CV

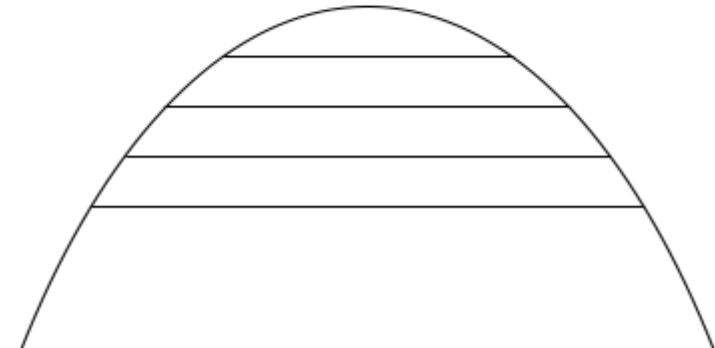
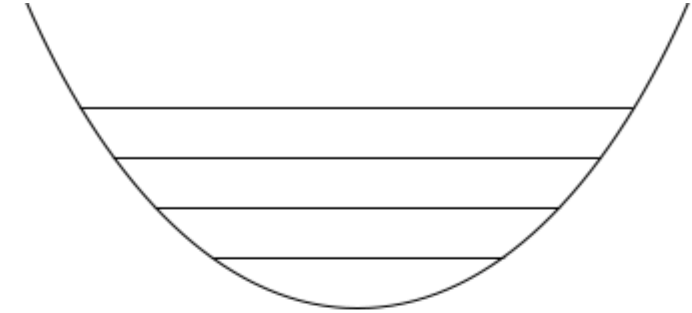
- Mercury Probe Contacts for C-V measurements
- The mercury contact does not damage the wafer nor leave mercury on the surface
- Contact either to the sample bottom or to the top



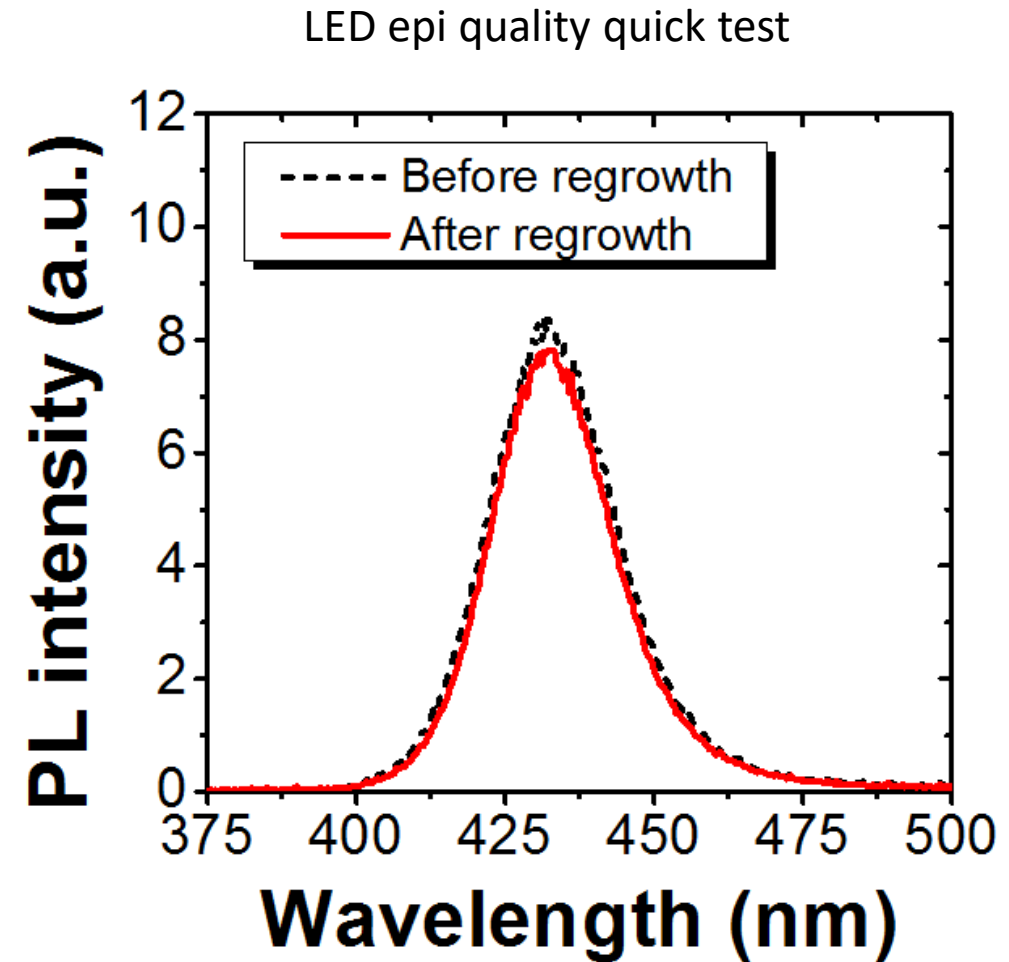
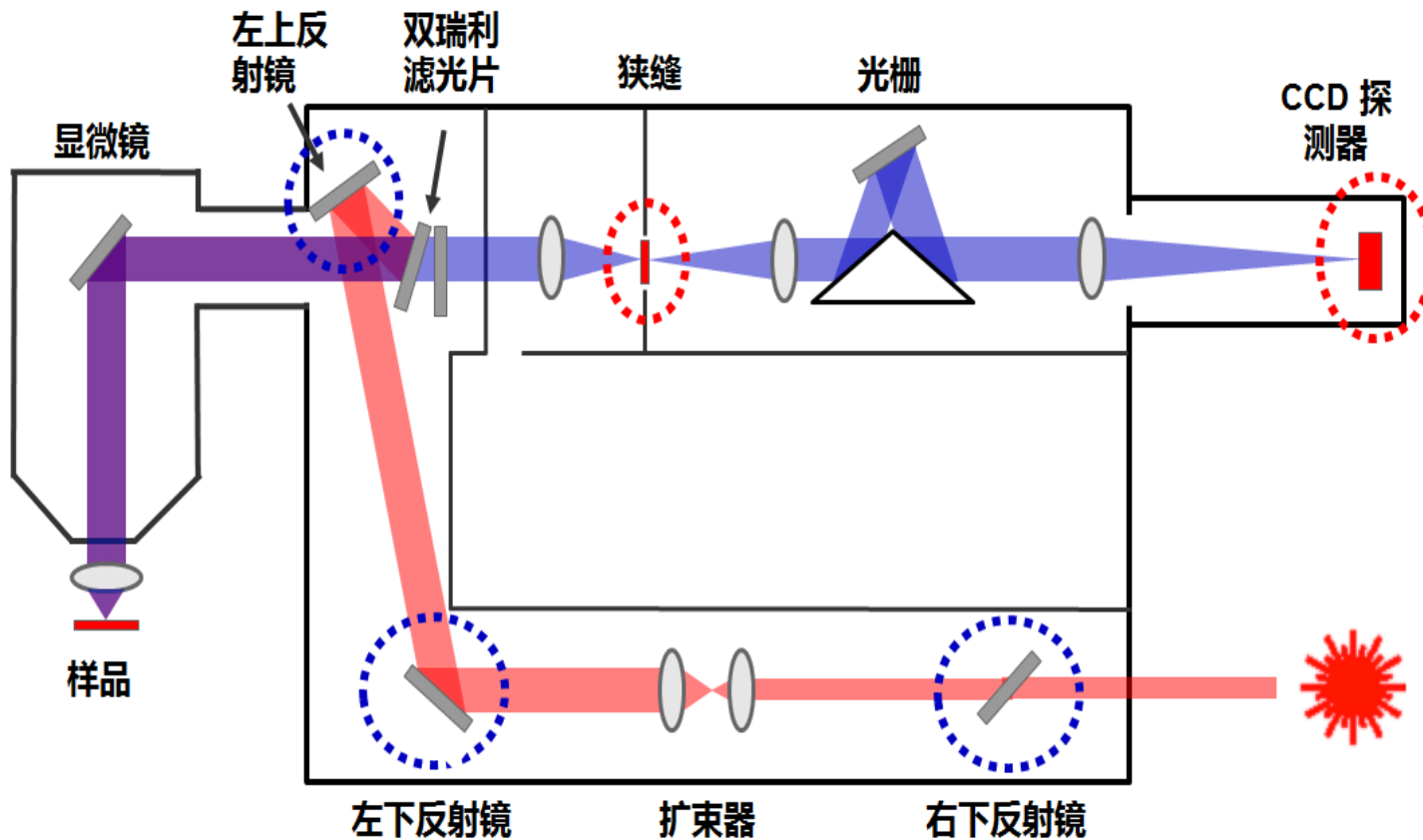
AlGaIn/GaN HEMT: CV → epi thickness & doping profile

PL

- Photoluminescence (PL)
- To detect and **identify impurities in semiconductor** materials by a beam of laser
- Creation of electron-hole pairs by incident radiation and subsequent radiative recombination photon emission



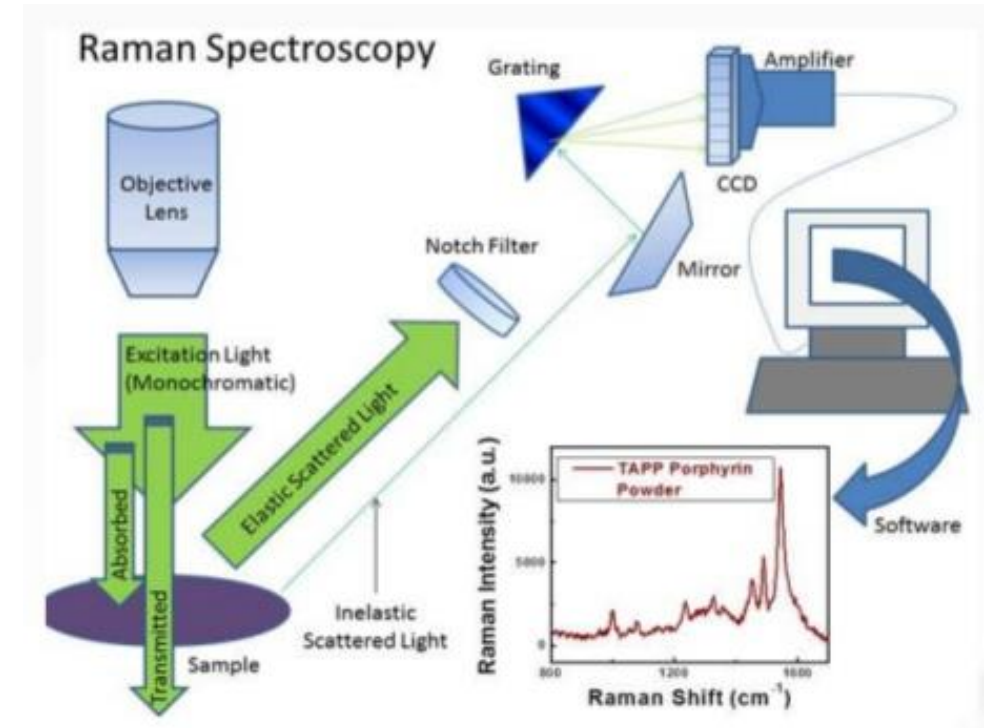
Renishaw Invia spectrometer



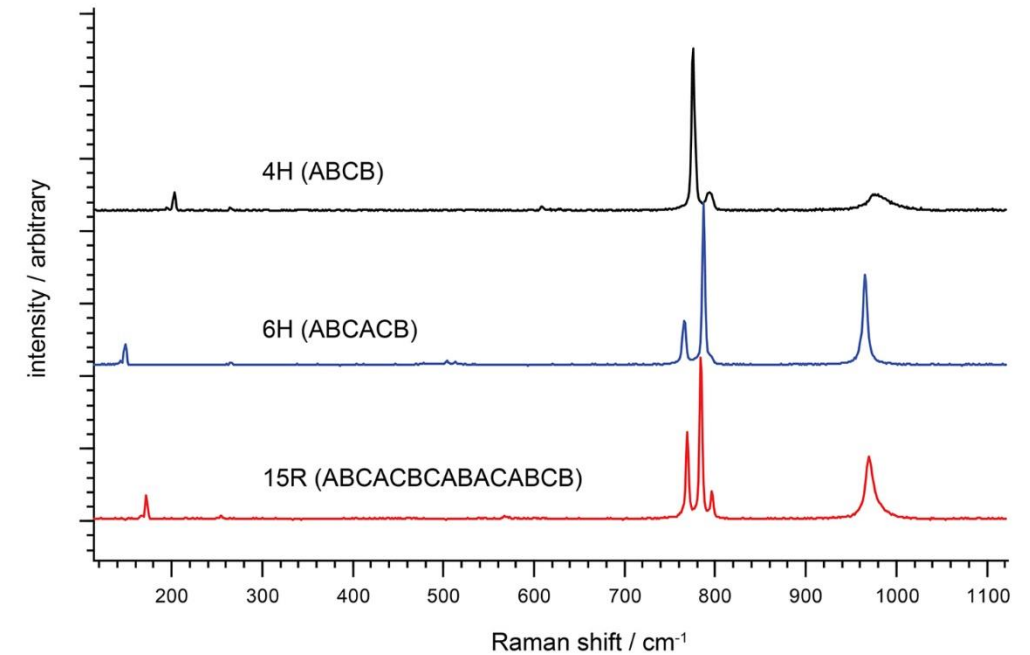
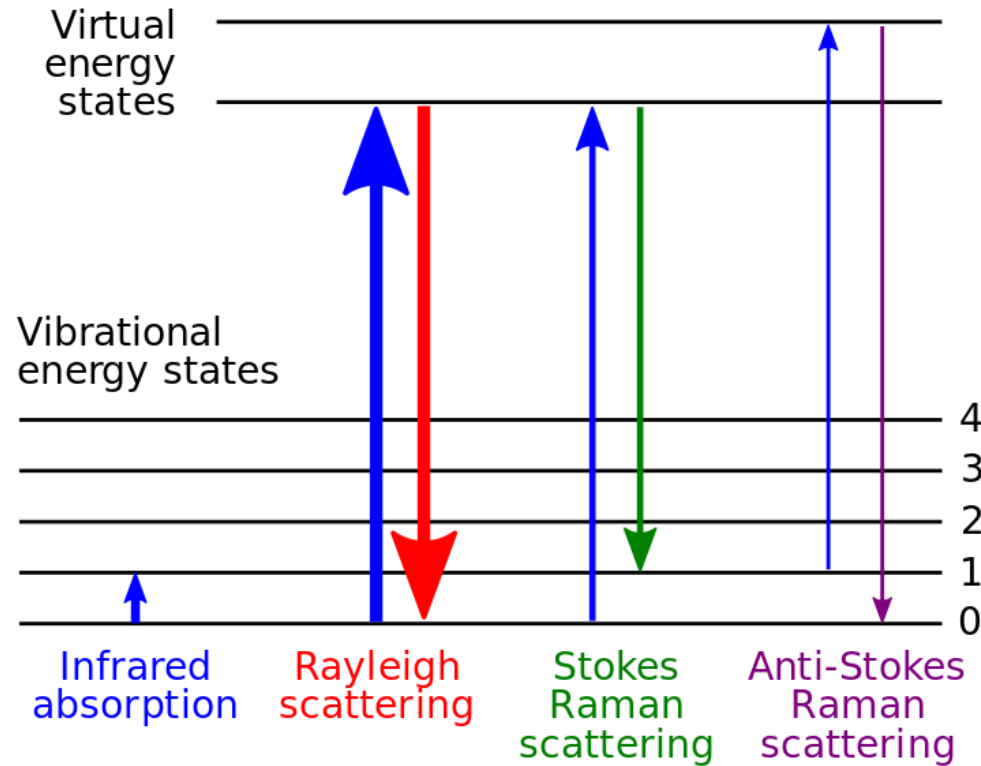
Raman spectroscopy

- Raman spectroscopy
- Detect both organic and inorganic species and measure the crystallinity of solids
- Sensitive to strain, but free from charging

Raman spectroscopy is commonly used in chemistry to provide a structural fingerprint by which molecules can be identified. It relies on inelastic scattering, or Raman scattering, of monochromatic light, usually from a laser in the visible, near infrared, or near ultraviolet range. The laser light interacts with molecular vibrations, phonons or other excitations in the system, resulting in the energy of the laser photons being shifted up or down. The shift in energy gives information about the vibrational modes in the system



Raman

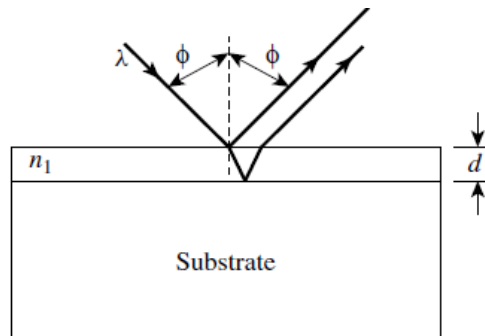


Spectra of three different polytypes of SiC

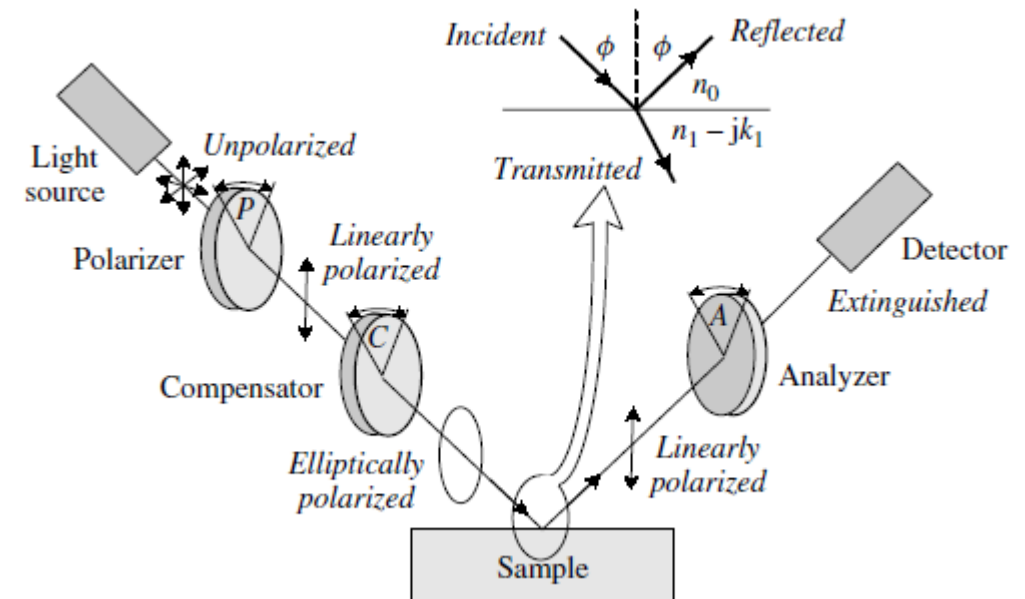
When light is scattered from the surface of a sample, the scattered light is found to contain mainly wavelengths that were incident on the sample (Rayleigh scattering) but also at different wavelengths at very low intensities (few parts per million or less) that represent an interaction of the incident light with the material.

Ellipsometry

- Ellipsometry
- To determine the thickness of thin dielectric films on absorbing substrates, line width, and optical constants of films or substrates



It has been also used to monitor the growth of superlattice structures grown by molecular beam epitaxy (MBE) and metalorganic chemical vapor deposition (MOCVD)





What you can take away from this course

- GaN epitaxy methods
- Characterization techniques of the third generation semiconductors
- Homework: classify all the characterization techniques in this chapter to THREE categories. Explain why you classify in this way.



Case study – Defect inspection of GaN materials

Candela 8720
MicroLED Bare Wafer Inspection Solution

Industry need: Macro & Micro Defect Classification

Full Wafer Mapping
Macro & Micro Defect Classification

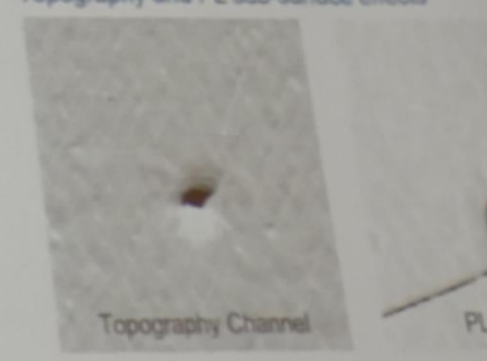
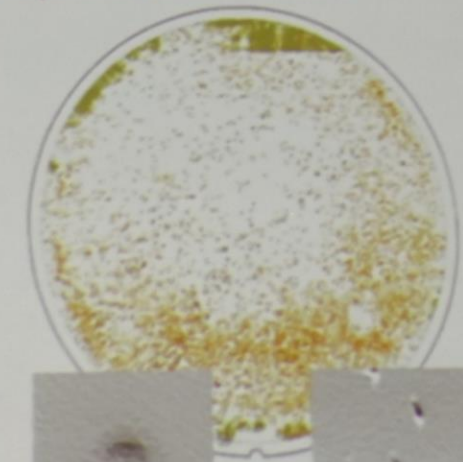
- Surface Defect Classification (Epi Defects, Hex Bumps & Hillocks)
- GaN Micropit vs. Particle Separation
- 0.06 μ m PSL on Si sensitivity

Industry need: System Capable of Multiple-Material Inspection

Multiple Material Capable
Material & Device Unique Defects

- GaN Cracks
- GaN on Sapphire PL defects
- GaN on SiC PL defects
- Other material like GaAs, InP, LiNbO₃/UT

Hillock Defect detected with Surface Topography and PL sub-surface effects

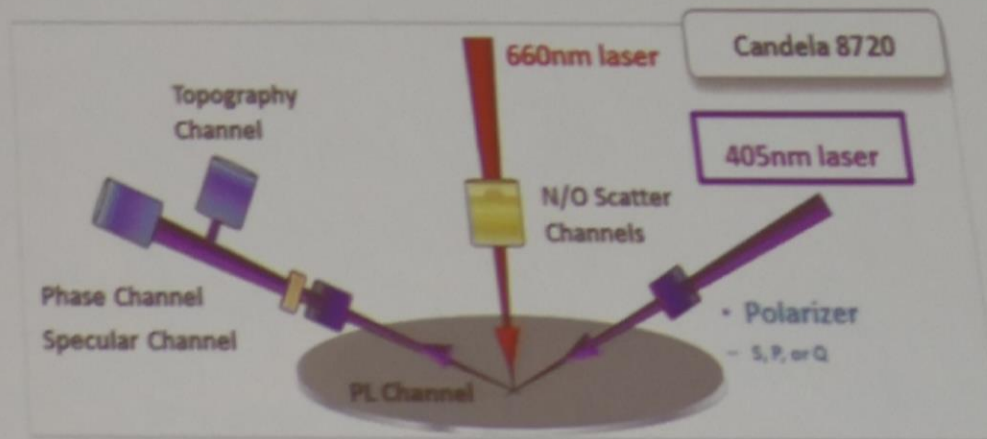


Case study – Defect inspection of GaN materials

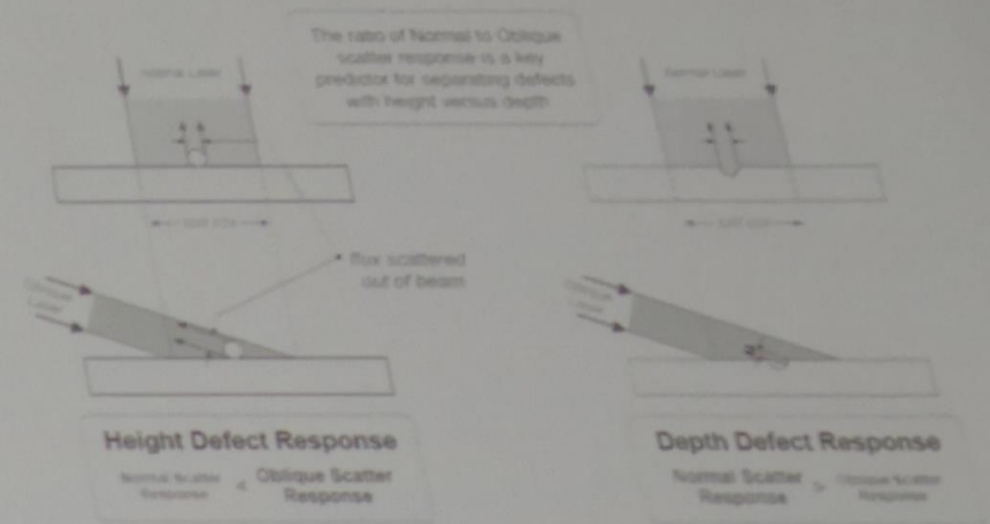
Candela 8720系统概览

Overview

Optics



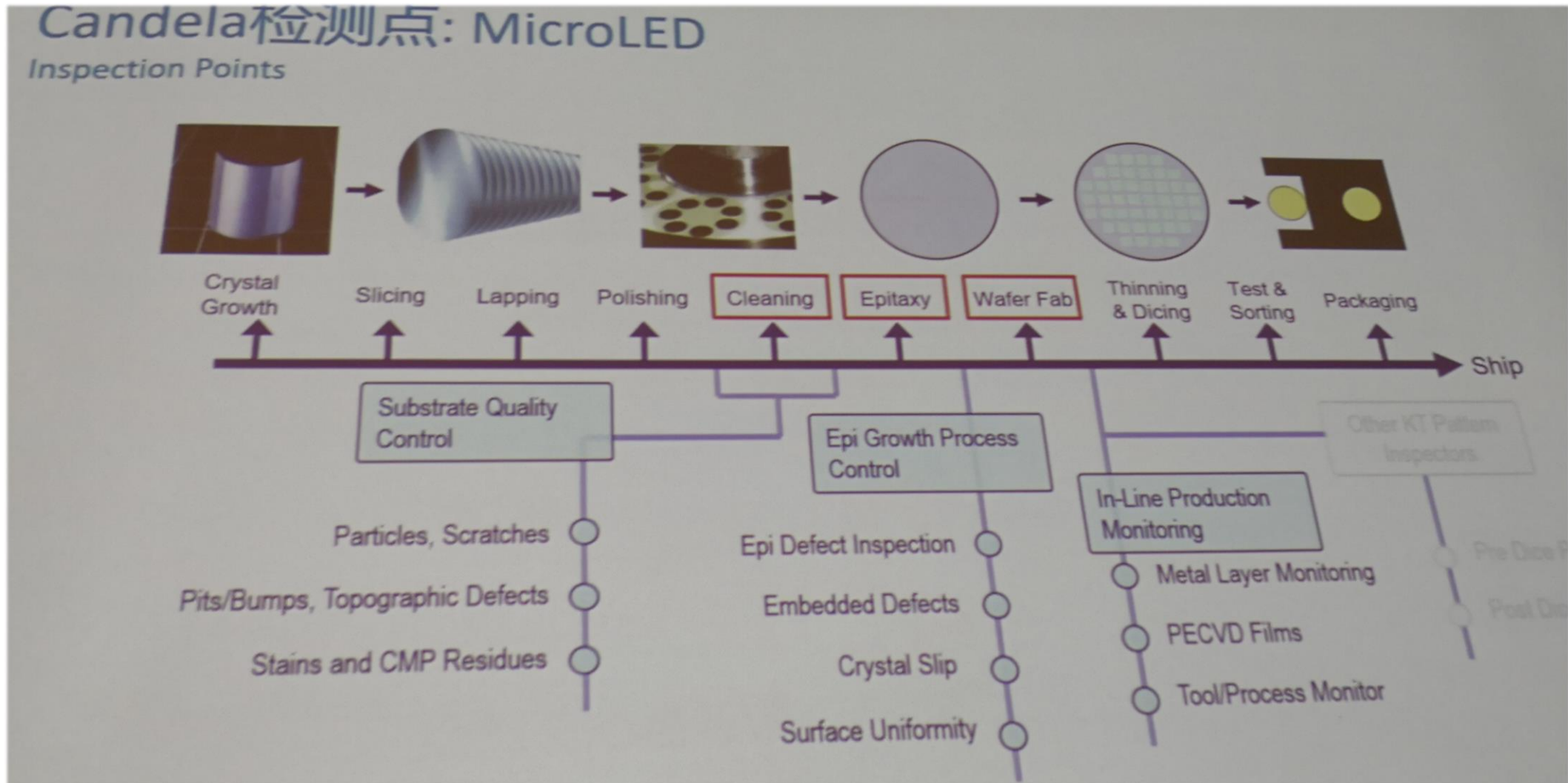
Pit/Particle Classification



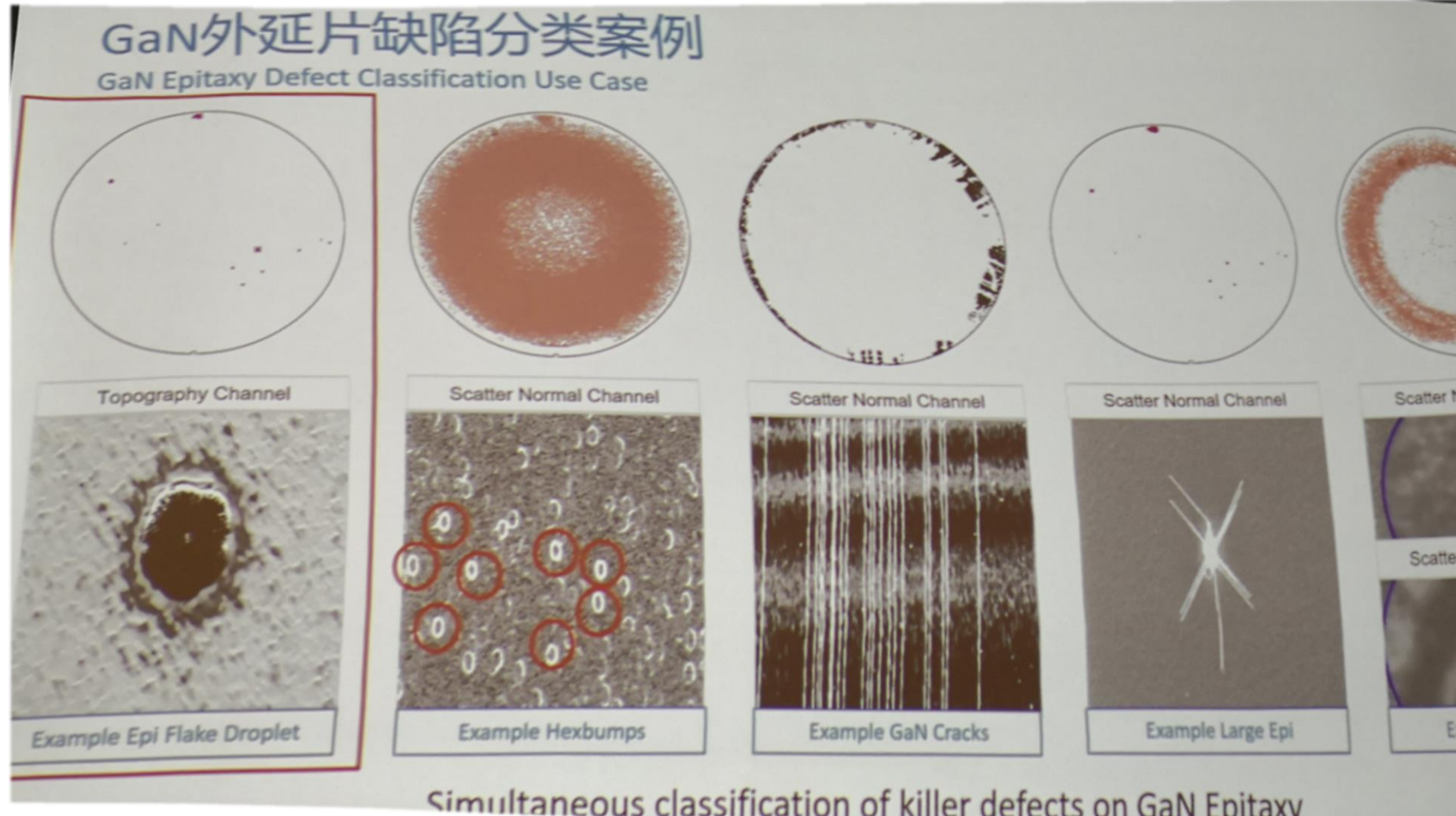
Data Acquisition

Inspection Flow

Case study – Defect inspection of GaN material & devices

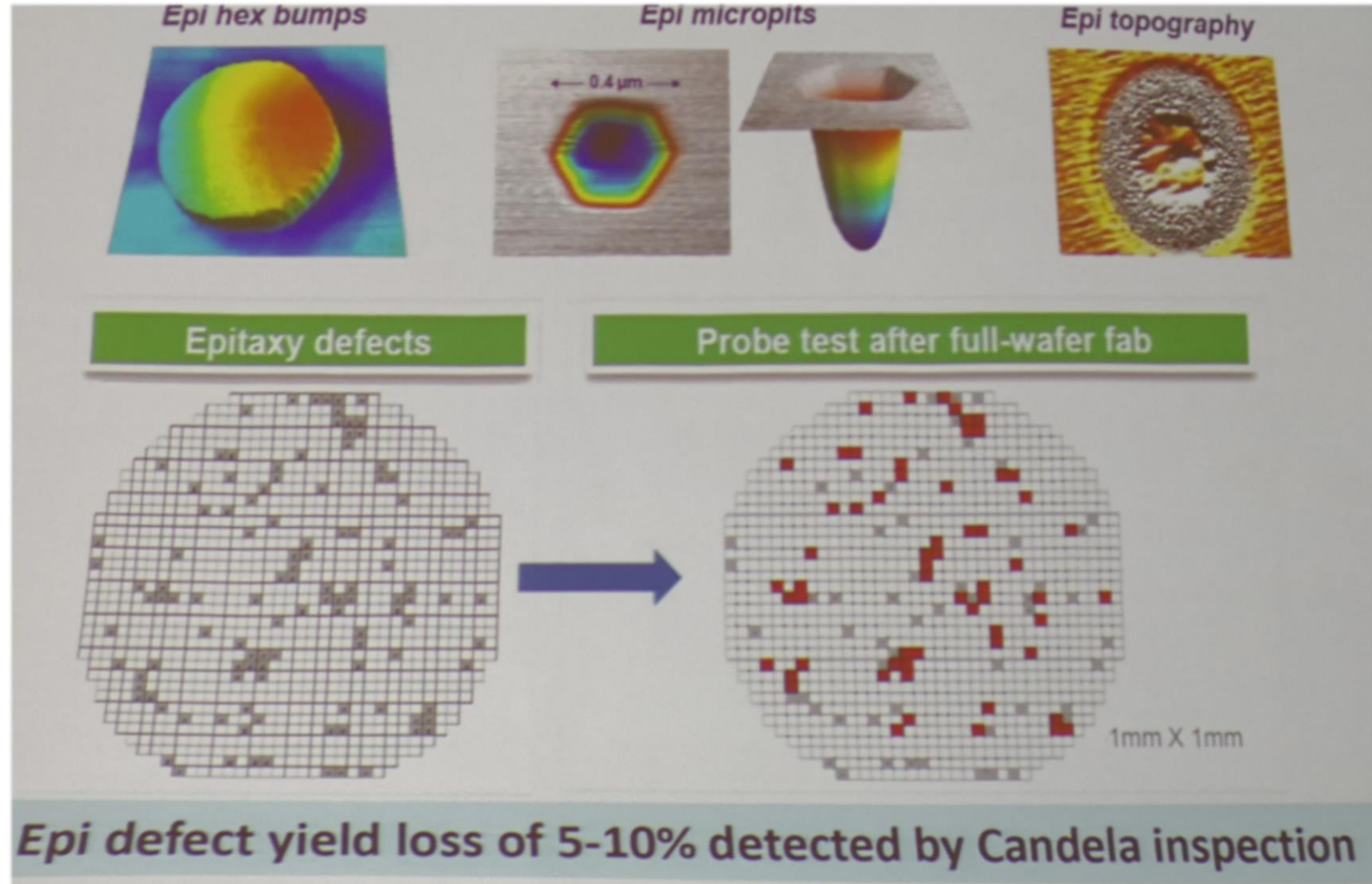


Case study – Defect inspection of GaN materials



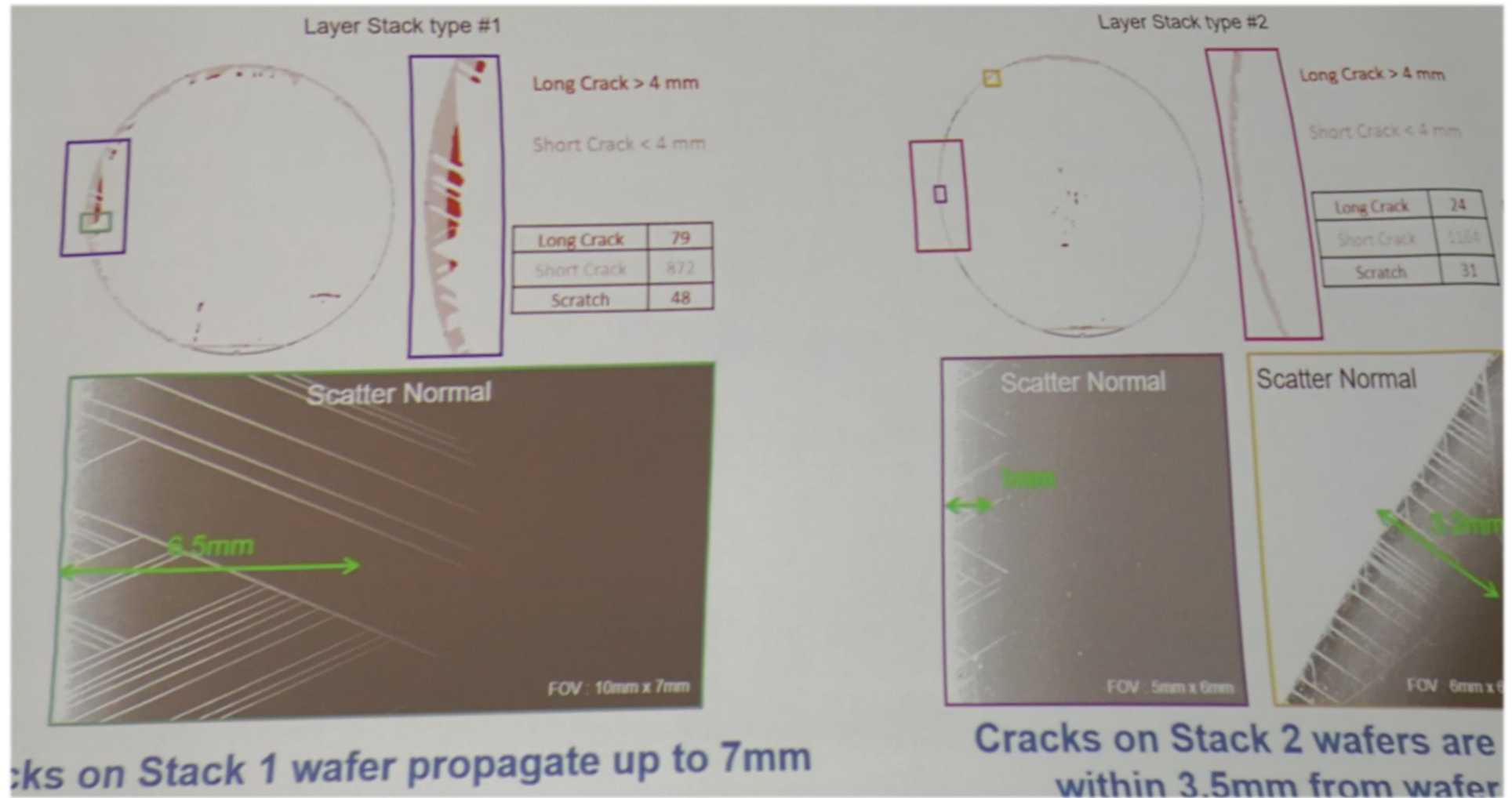
Case study – Defect inspection of GaN materials

外延工艺缺陷影响
最终器件良率



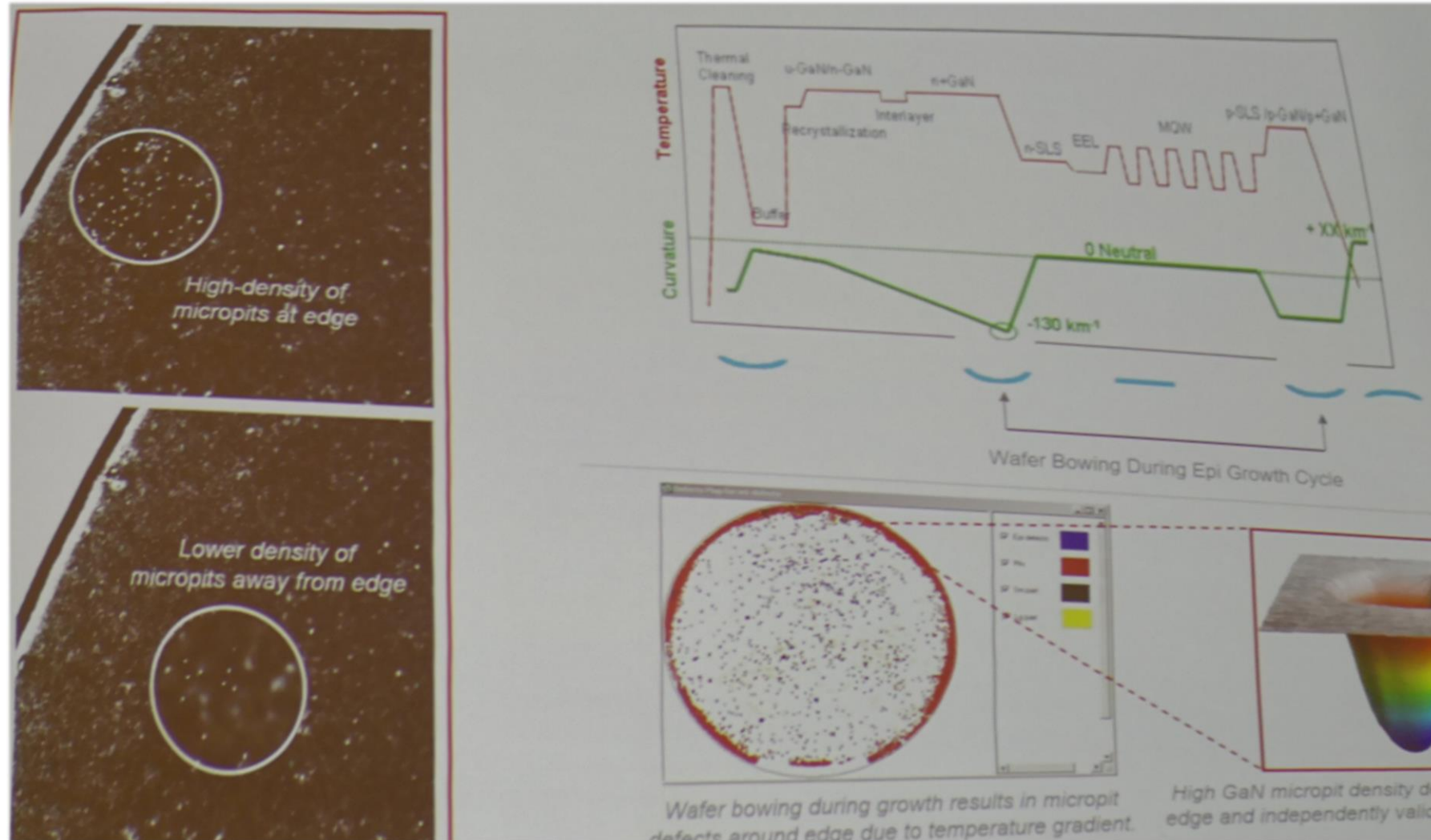
Case study – Defect inspection of GaN materials

不同外延层结构中裂纹缺陷的监测案例

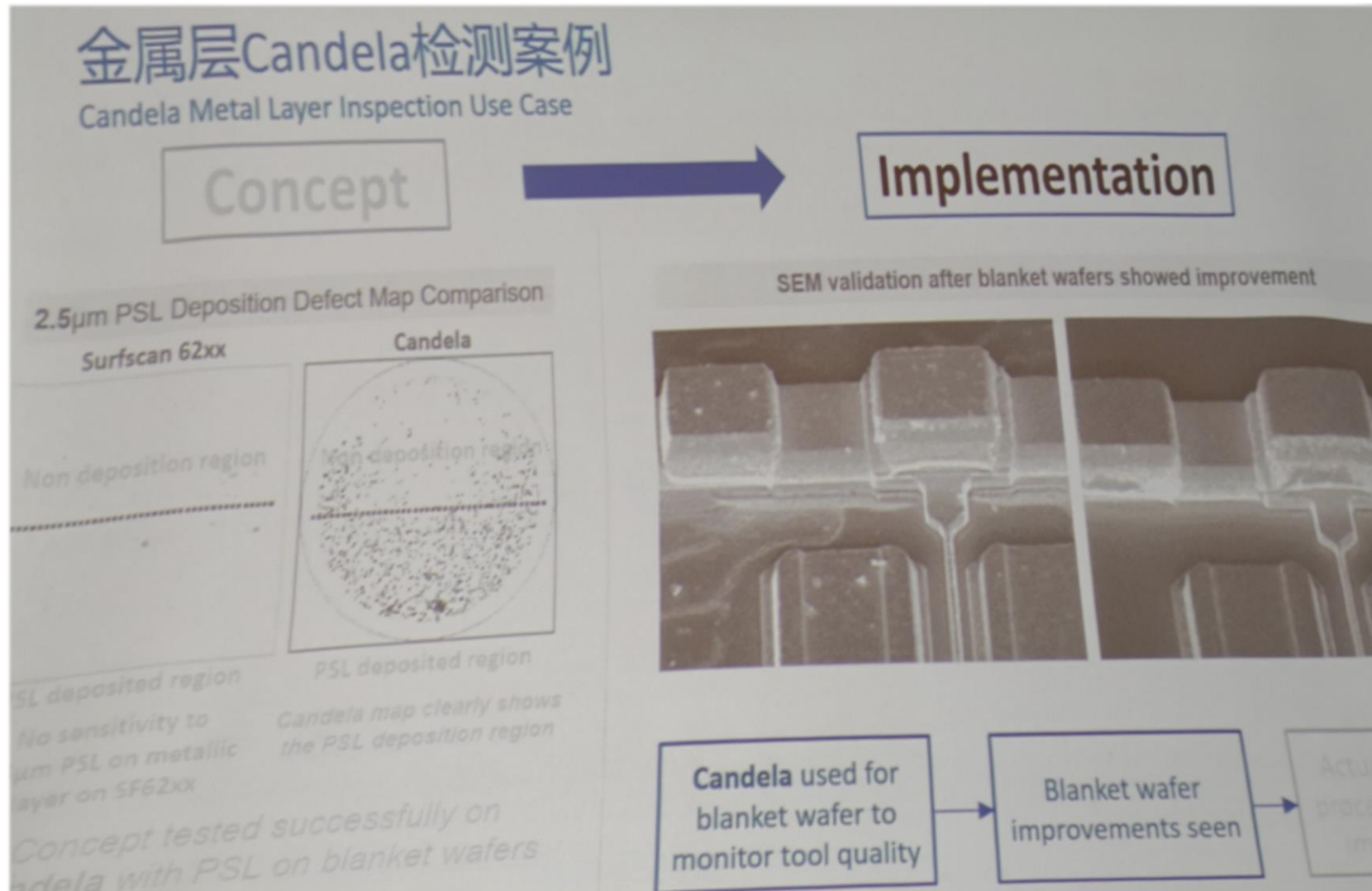


Case study – Defect inspection of GaN materials

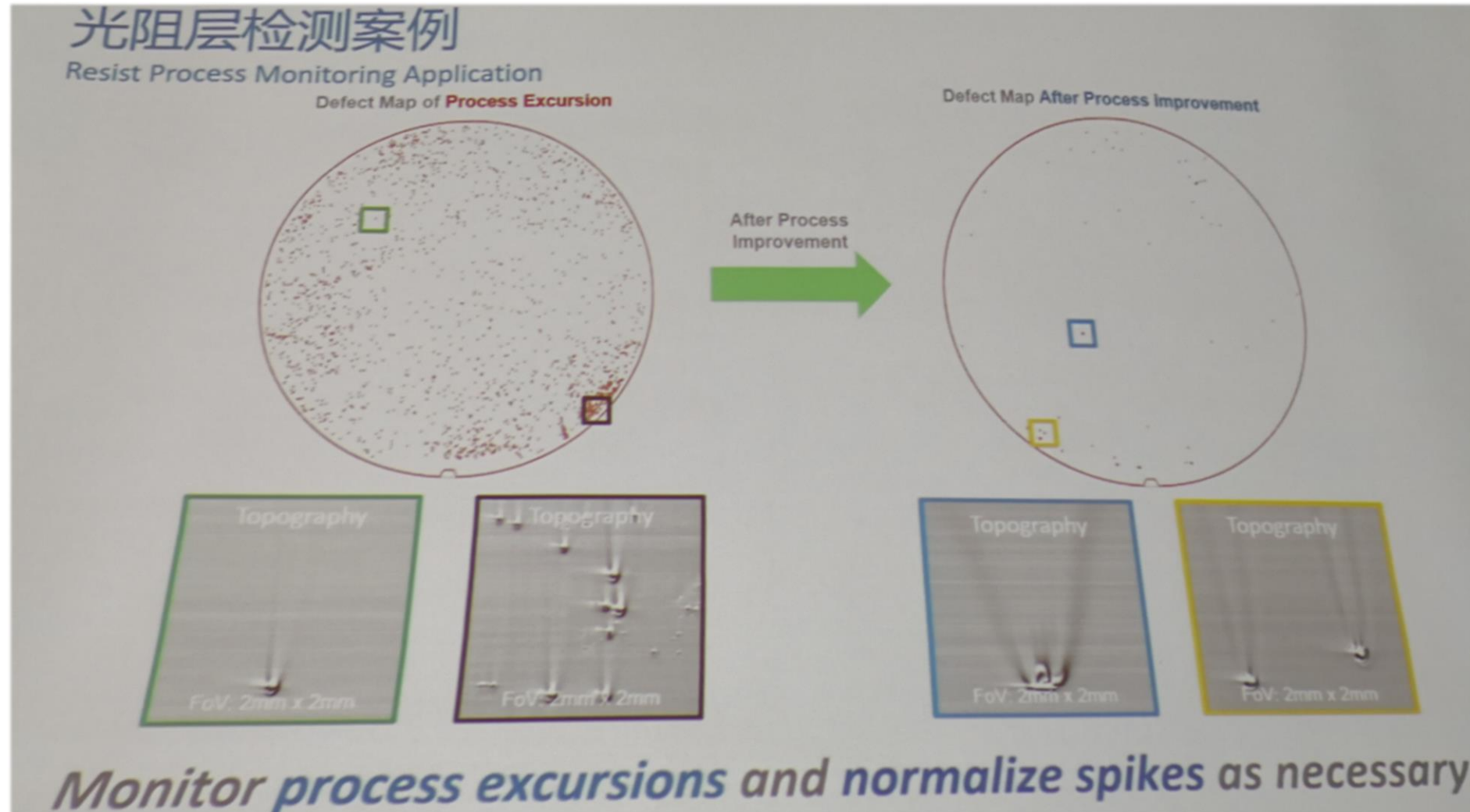
Micropit 验证
Reference Micropit
Validation



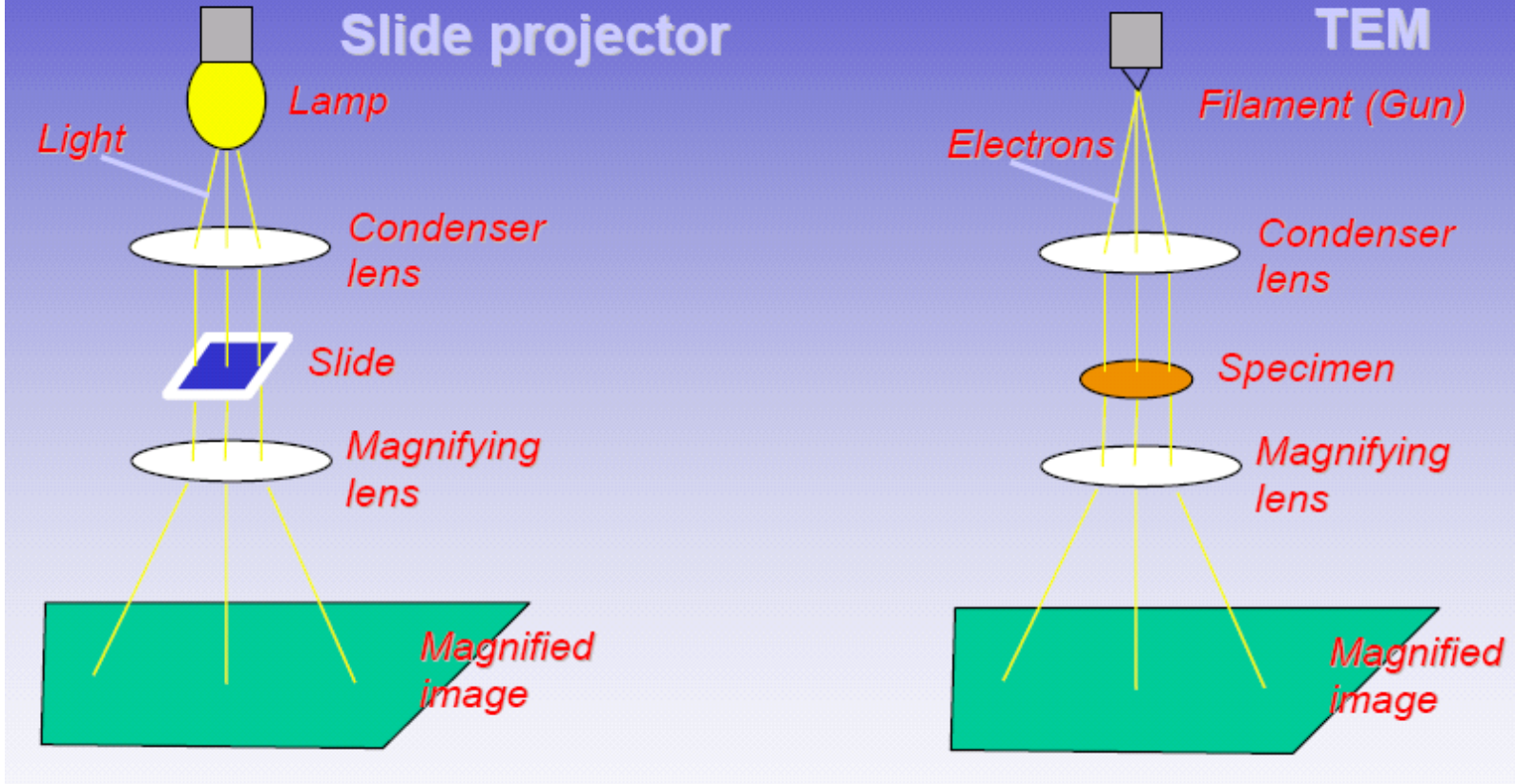
Case study – Defect inspection of GaN materials



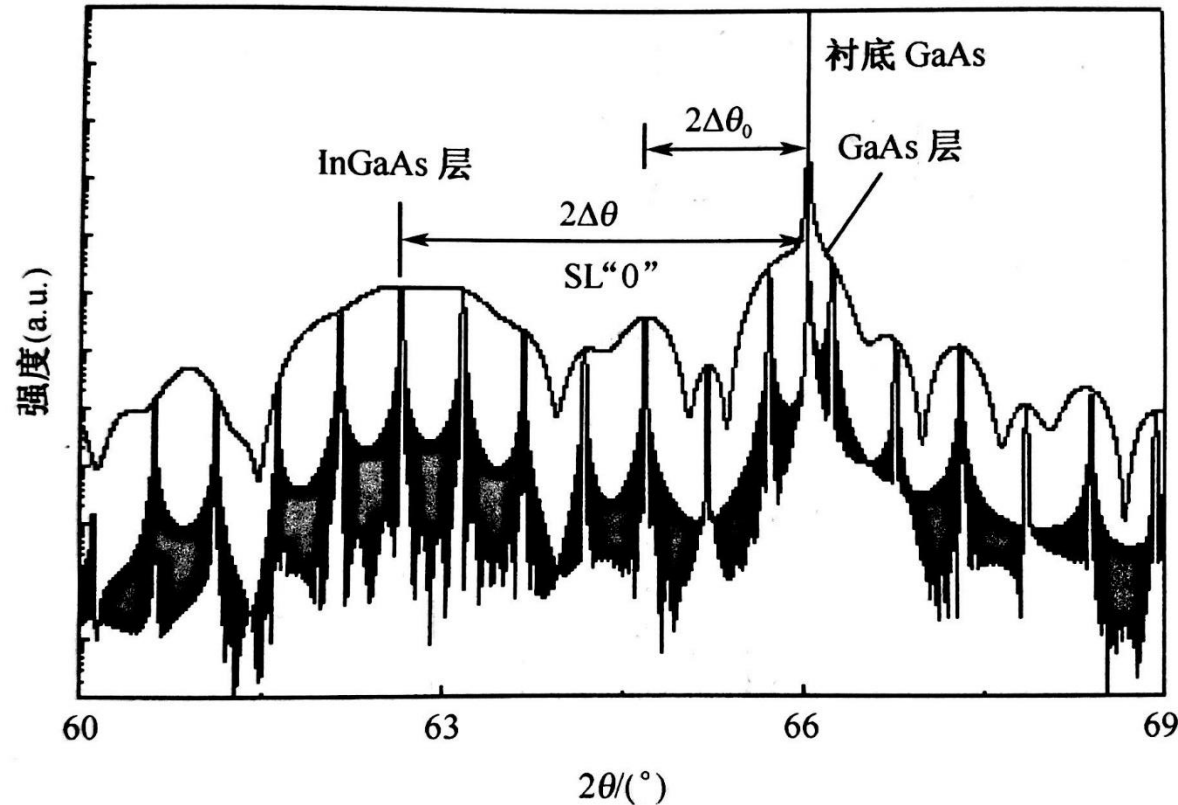
Case study – Defect inspection of GaN materials



Optical analogy



$[(\text{In}_x\text{Ga}_{1-x}\text{As})(\text{GaAs})]_M/\text{GaAs}$



Substrate: **GaAs**

Quantum Well: **InGaAs/GaAs**

periodic number: **M**

In composition: **X**

Ga composition: **1-X**

$\Delta\theta_0$: the distance between the zero order diffraction peak and the substrate diffraction peak, represent the average composition of the MQWs.

$\Delta\theta$: the distance between the InGaAs diffraction peak and the substrate diffraction peak, represent the In composition X.

图 2-108 $[(\text{In}_{0.5}\text{Ga}_{0.5}\text{As})_l(\text{GaAs})_m]_{20}/\text{GaAs}(001)$ 超晶格的卫星峰与卫星峰的包络线

其中 $Q=004$, InGaAs 层厚 $t_a=50\text{\AA}$, GaAs 层厚 $t_b=100\text{\AA}$, 周期数 $M=20$